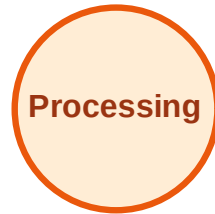


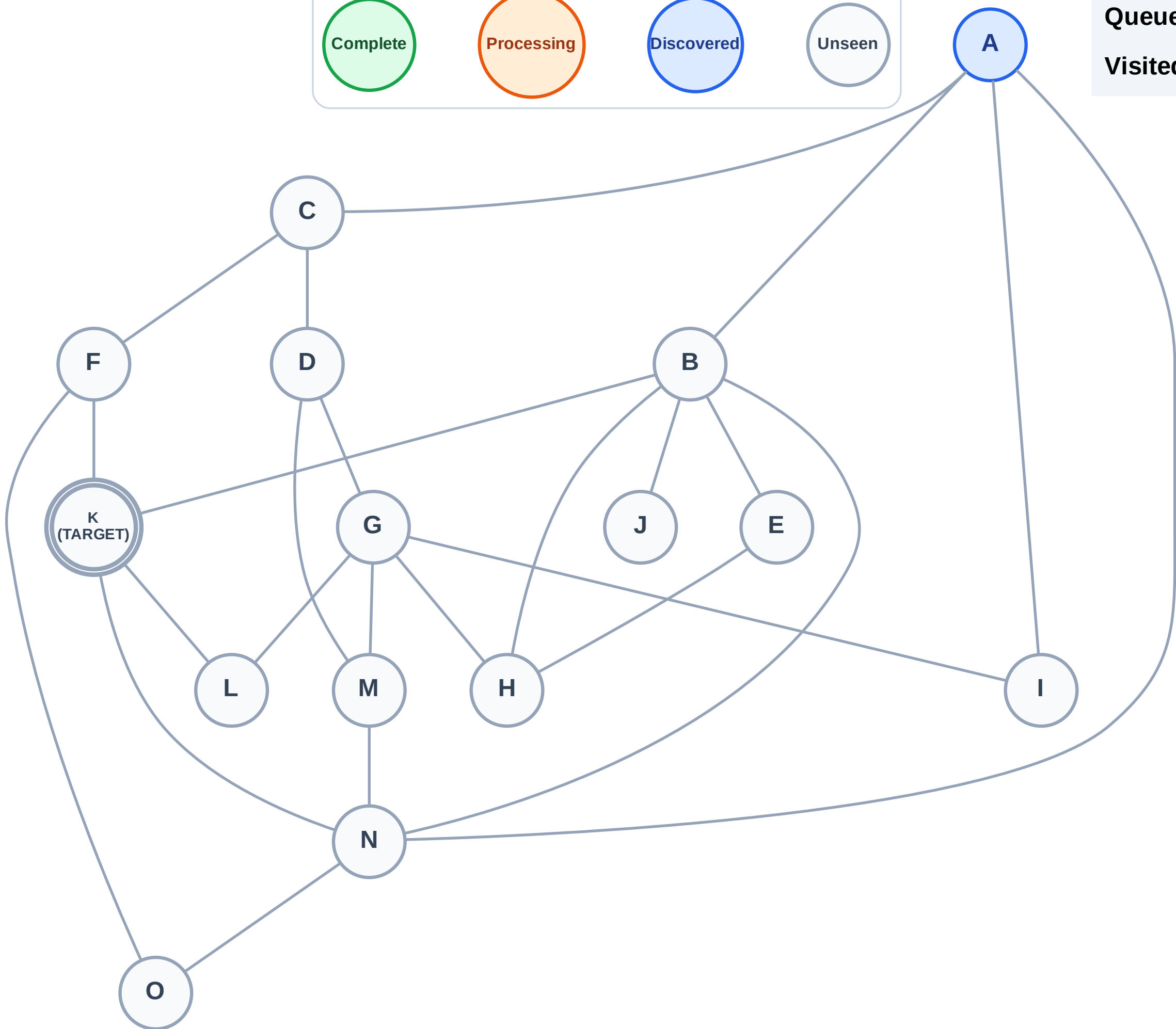
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



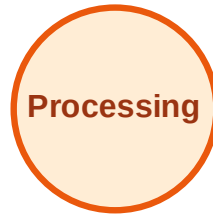
Queue: A
Visited: (none)



BFS Traversal

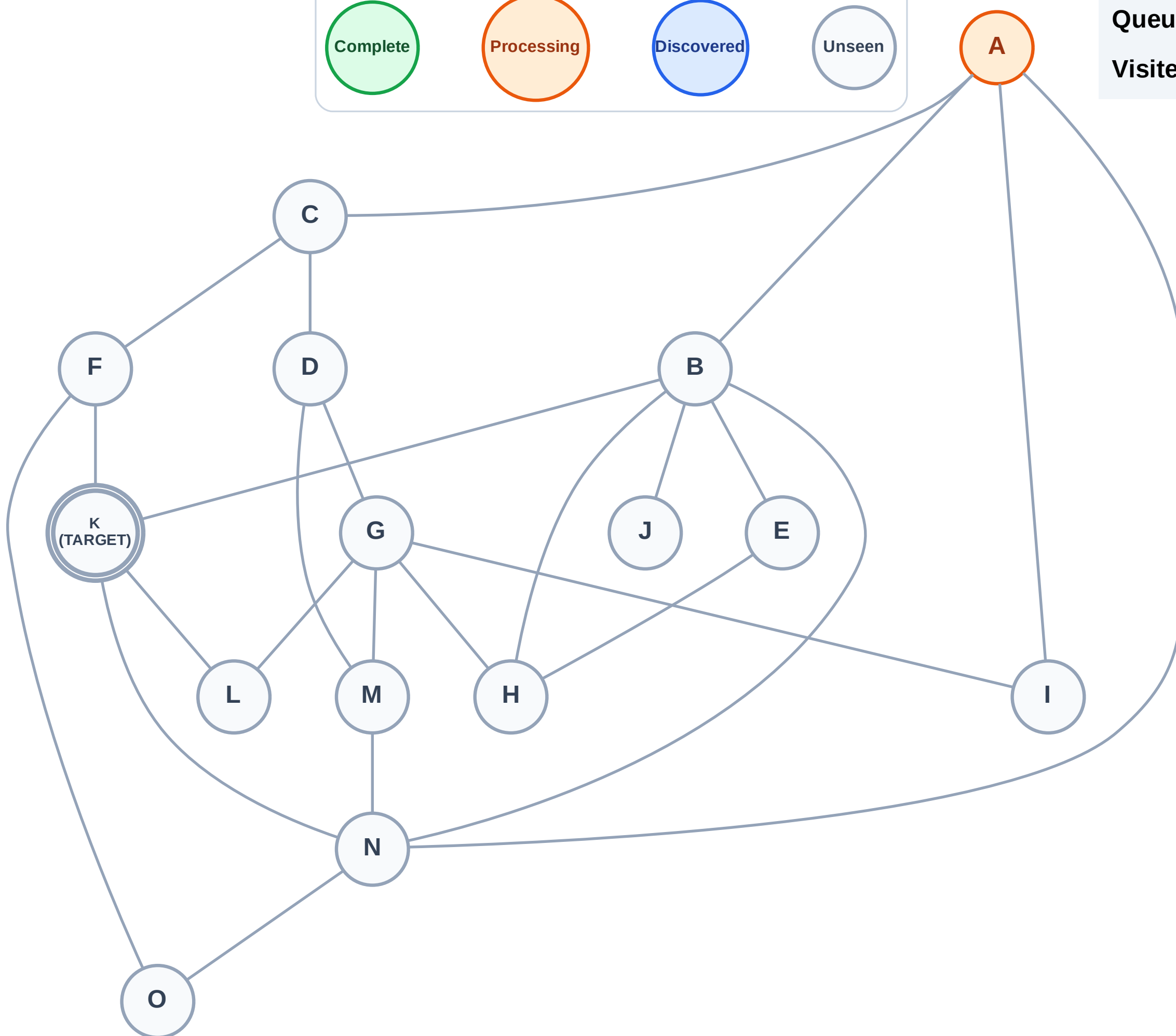
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



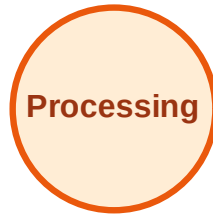
BFS Traversal

Step 3: Discover B, C, I, N from A

Legend



Complete



Processing

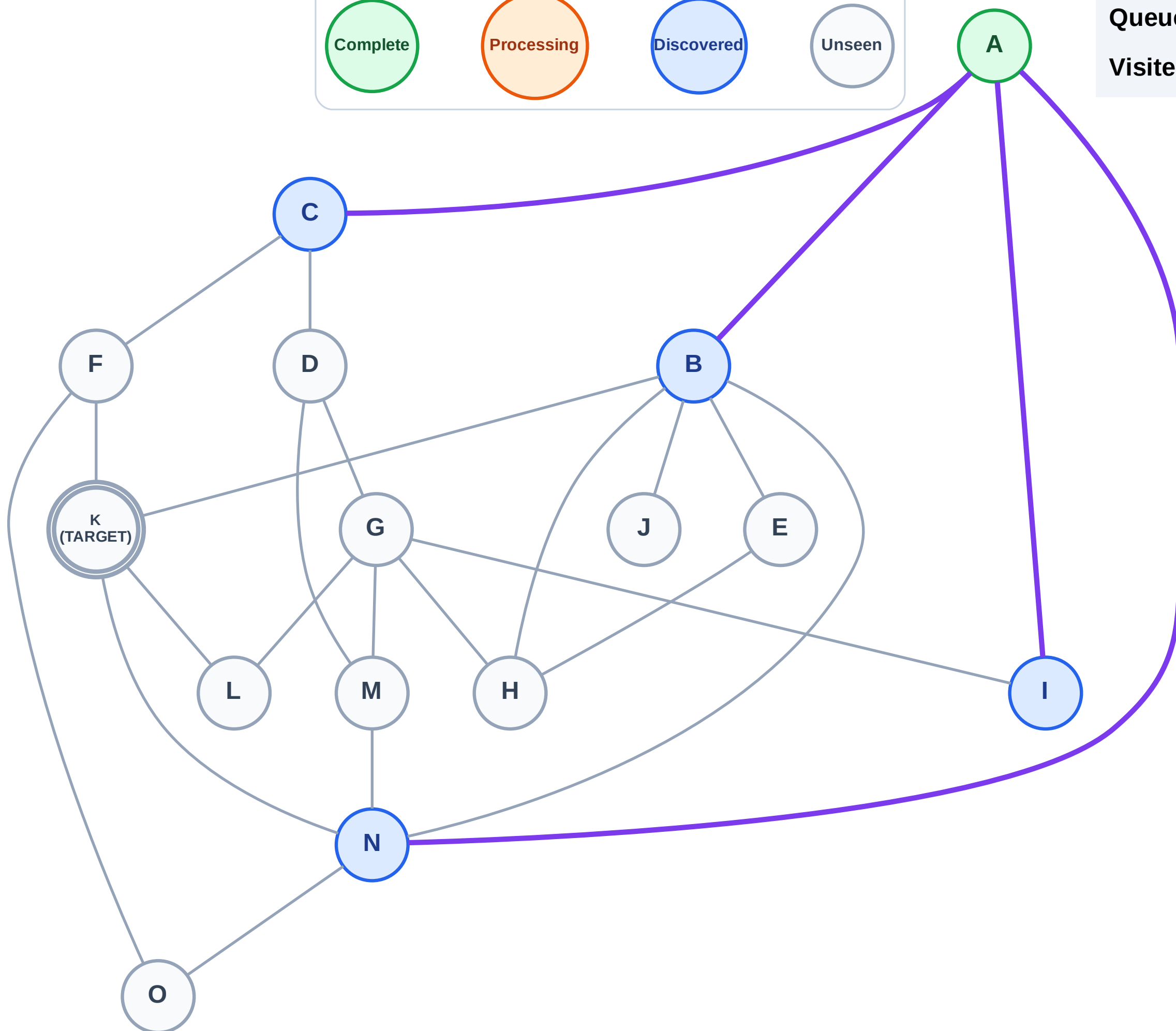


Discovered



Unseen

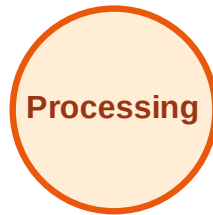
Queue: B → C → I → N
Visited: A



BFS Traversal

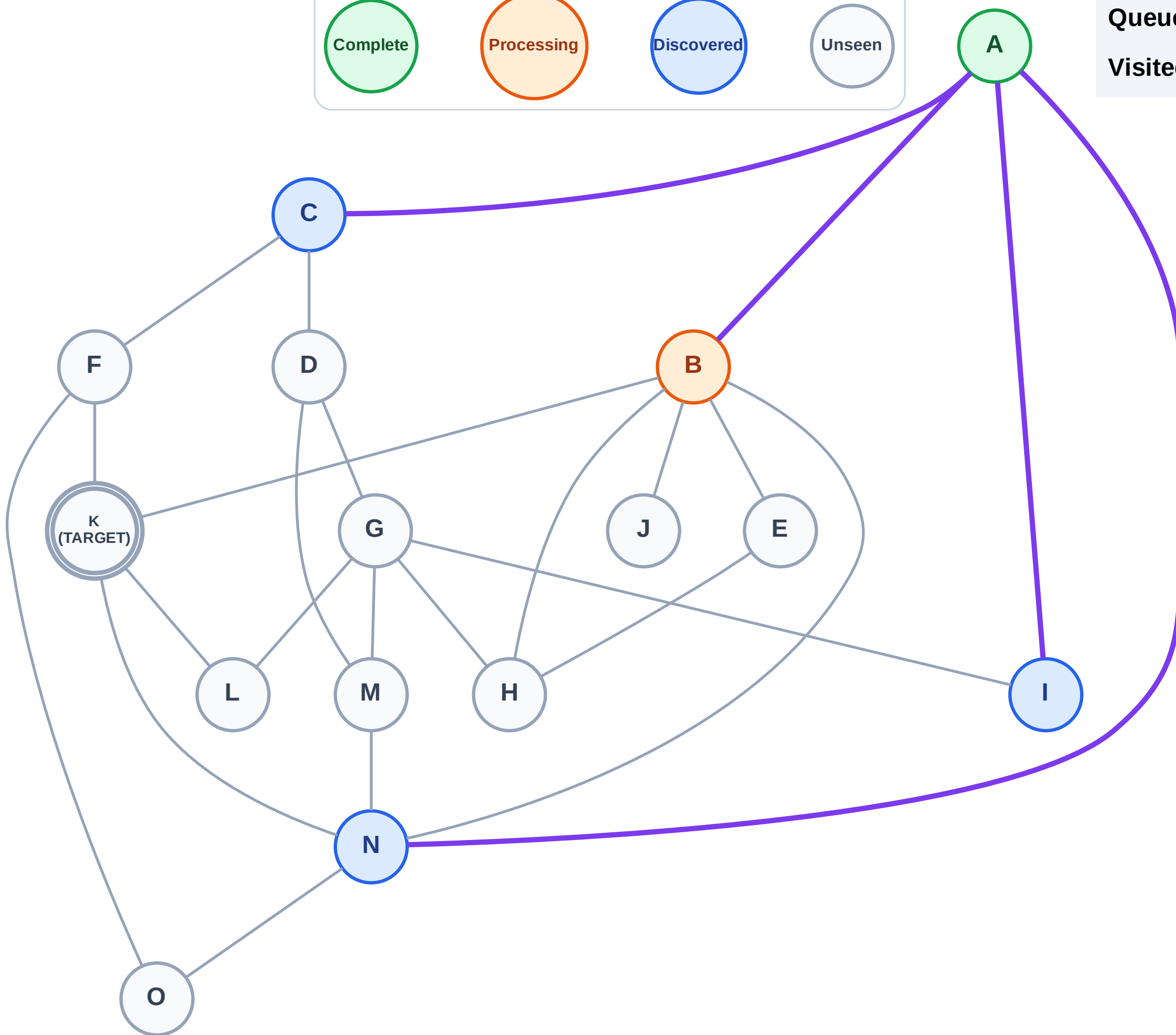
Step 4: Process node B

Legend



Queue: C → I → N

Visited: A



BFS Traversal

Step 5: Discover E, H, J, K from B

Legend

Complete

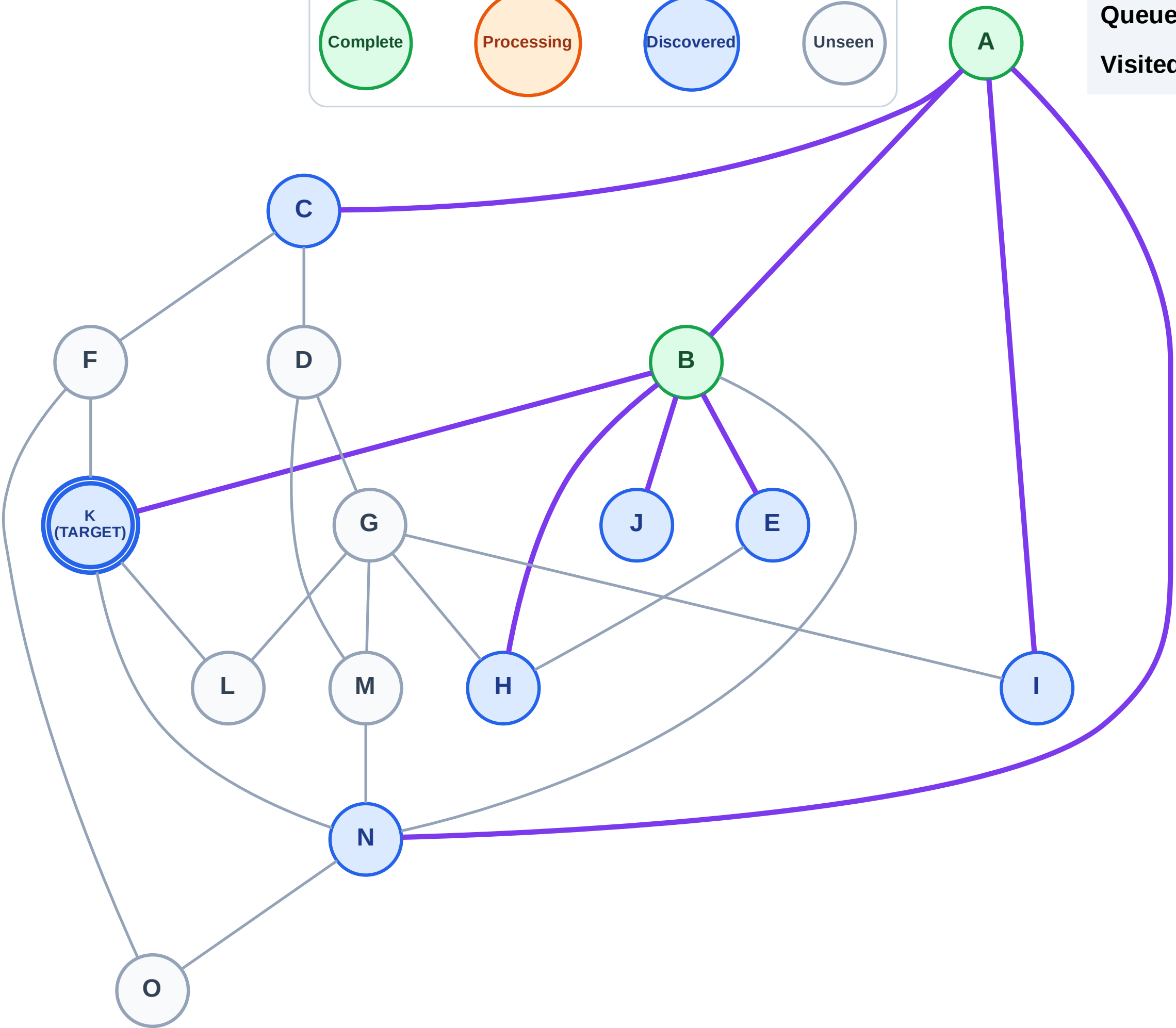
Processing

Discovered

Unseen

Queue: C → I → N → E → H → J → K

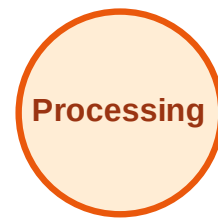
Visited: A, B



BFS Traversal

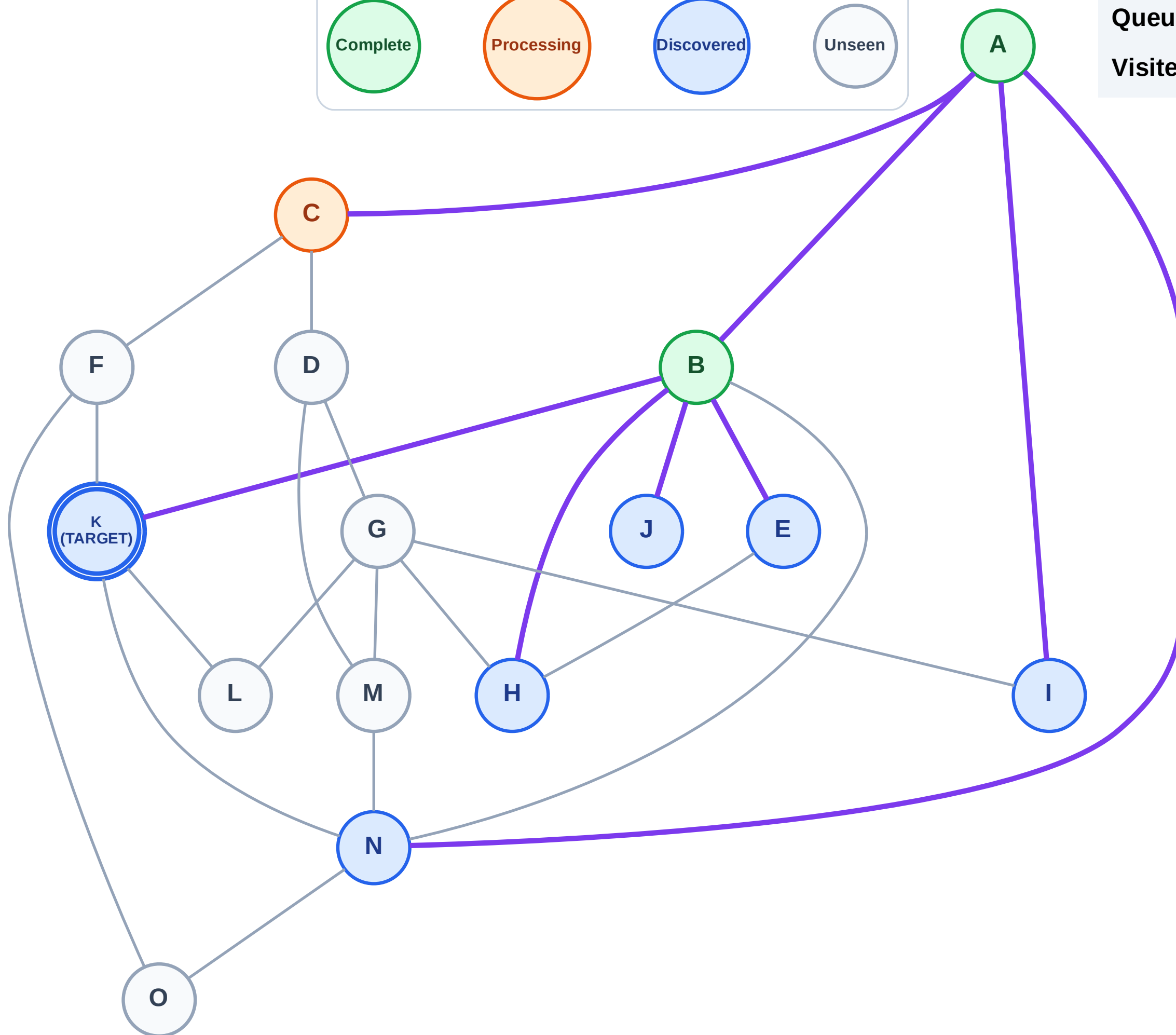
Step 6: Process node C

Legend



Queue: I → N → E → H → J → K

Visited: A, B



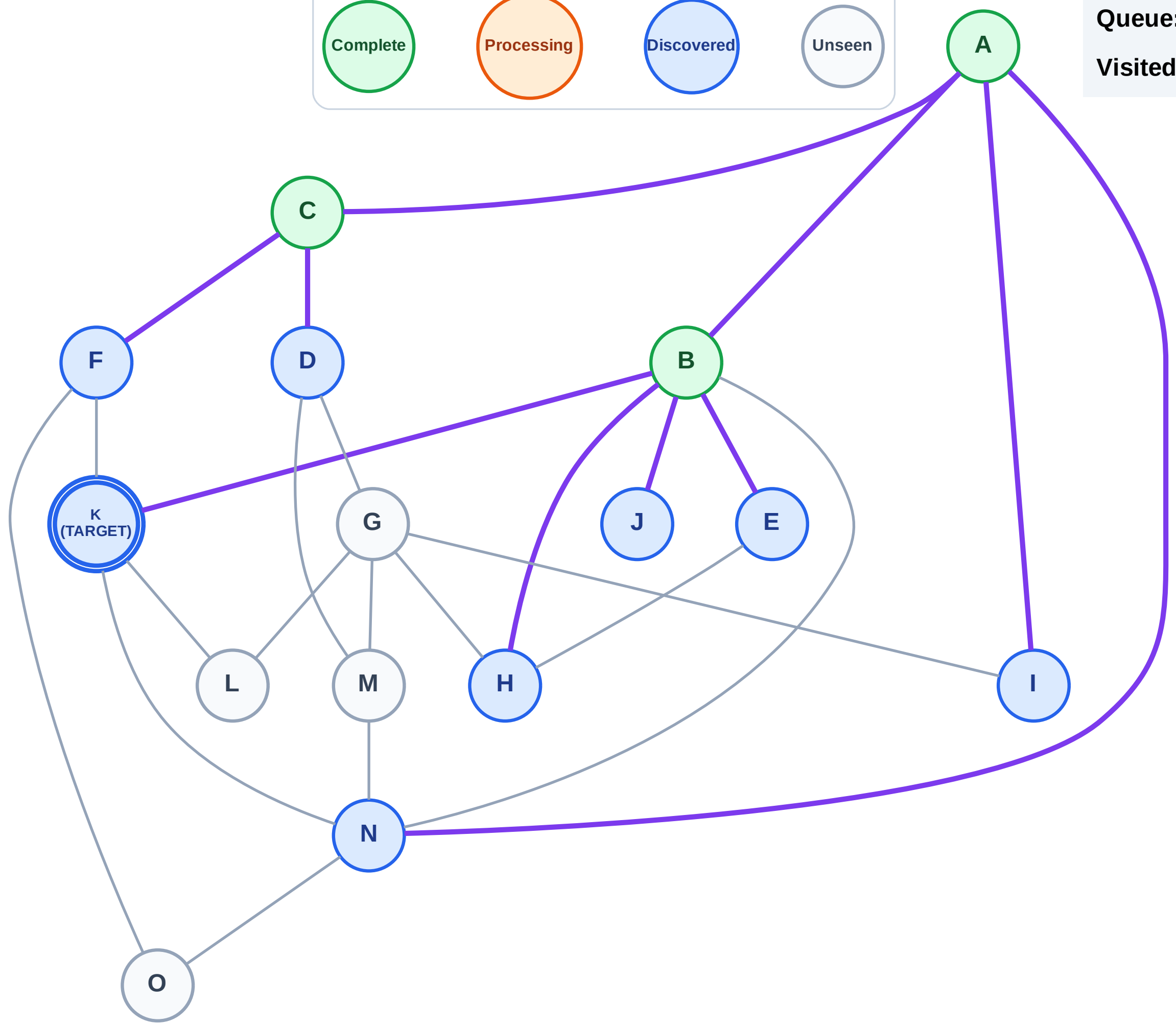
BFS Traversal

Step 7: Discover D, F from C

Legend



Queue: I → N → E → H → J → K → D → F
Visited: A, B, C



BFS Traversal

Step 8: Process node I

Legend

Complete

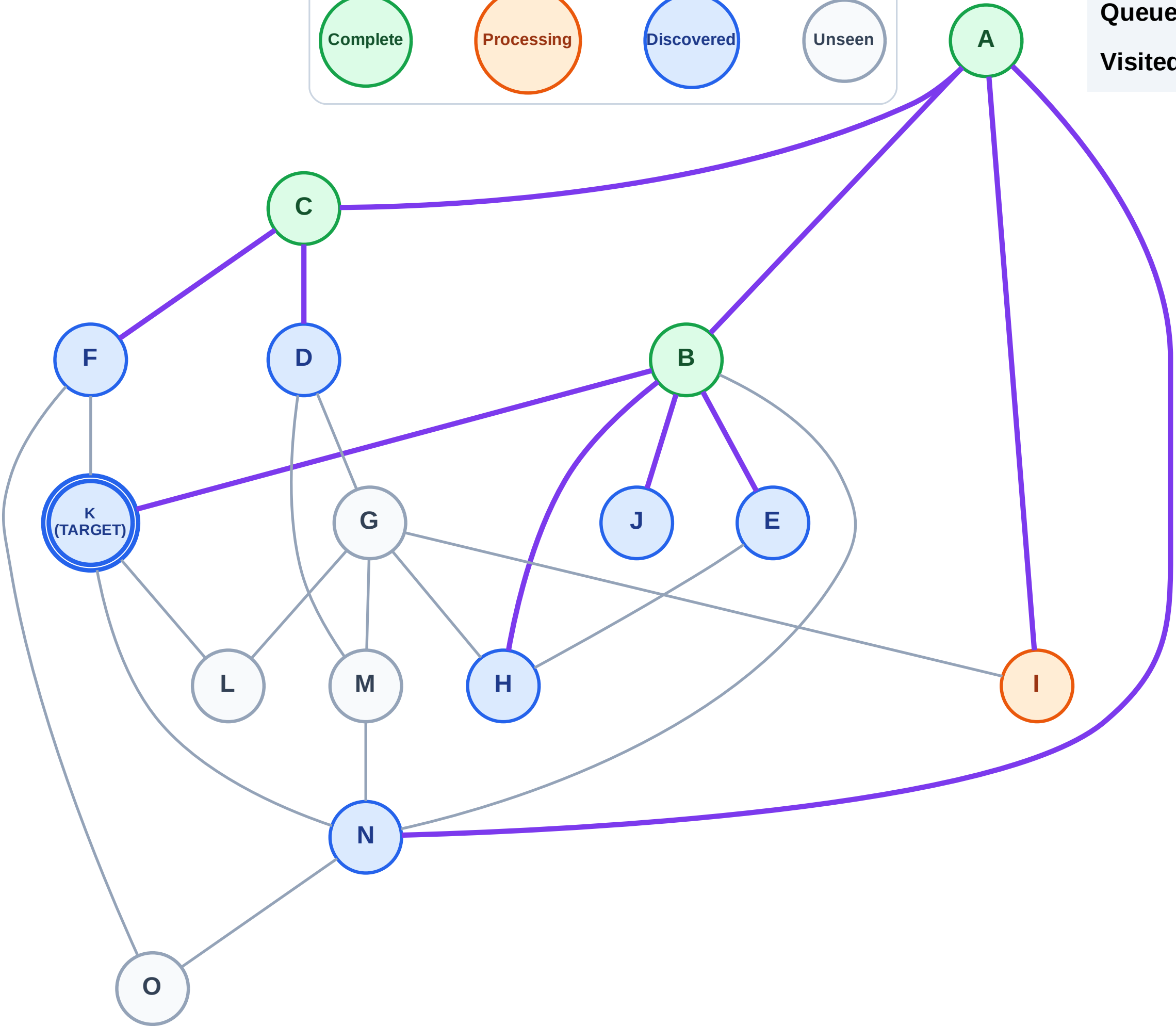
Processing

Discovered

Unseen

Queue: N → E → H → J → K → D → F

Visited: A, B, C



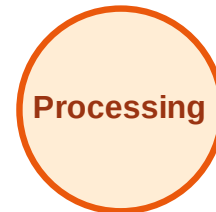
BFS Traversal

Step 9: Discover G from I

Legend



Complete



Processing



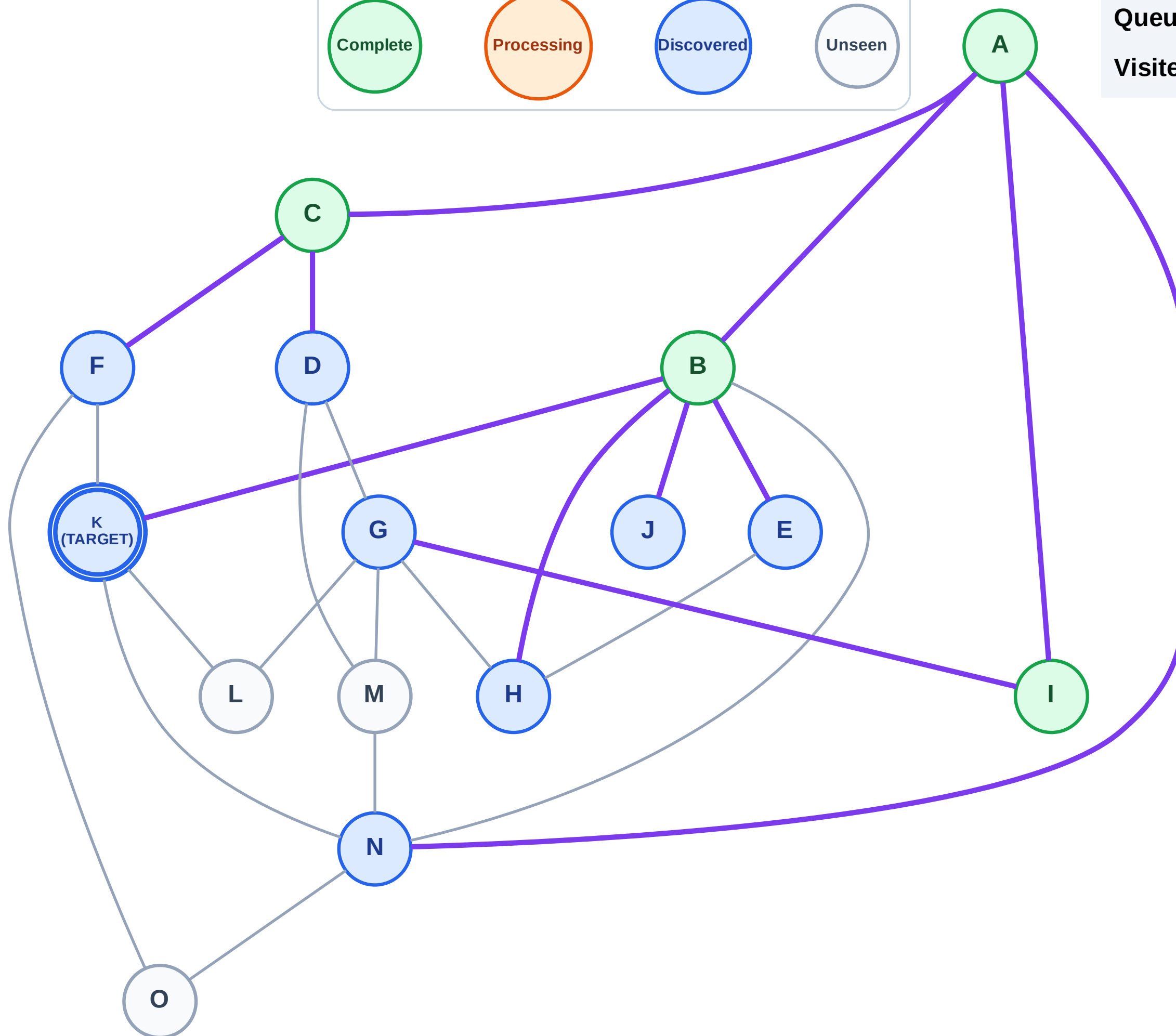
Discovered



Unseen

Queue: N → E → H → J → K → D → F → G

Visited: A, B, C, I



BFS Traversal

Step 10: Process node N

Legend

Complete

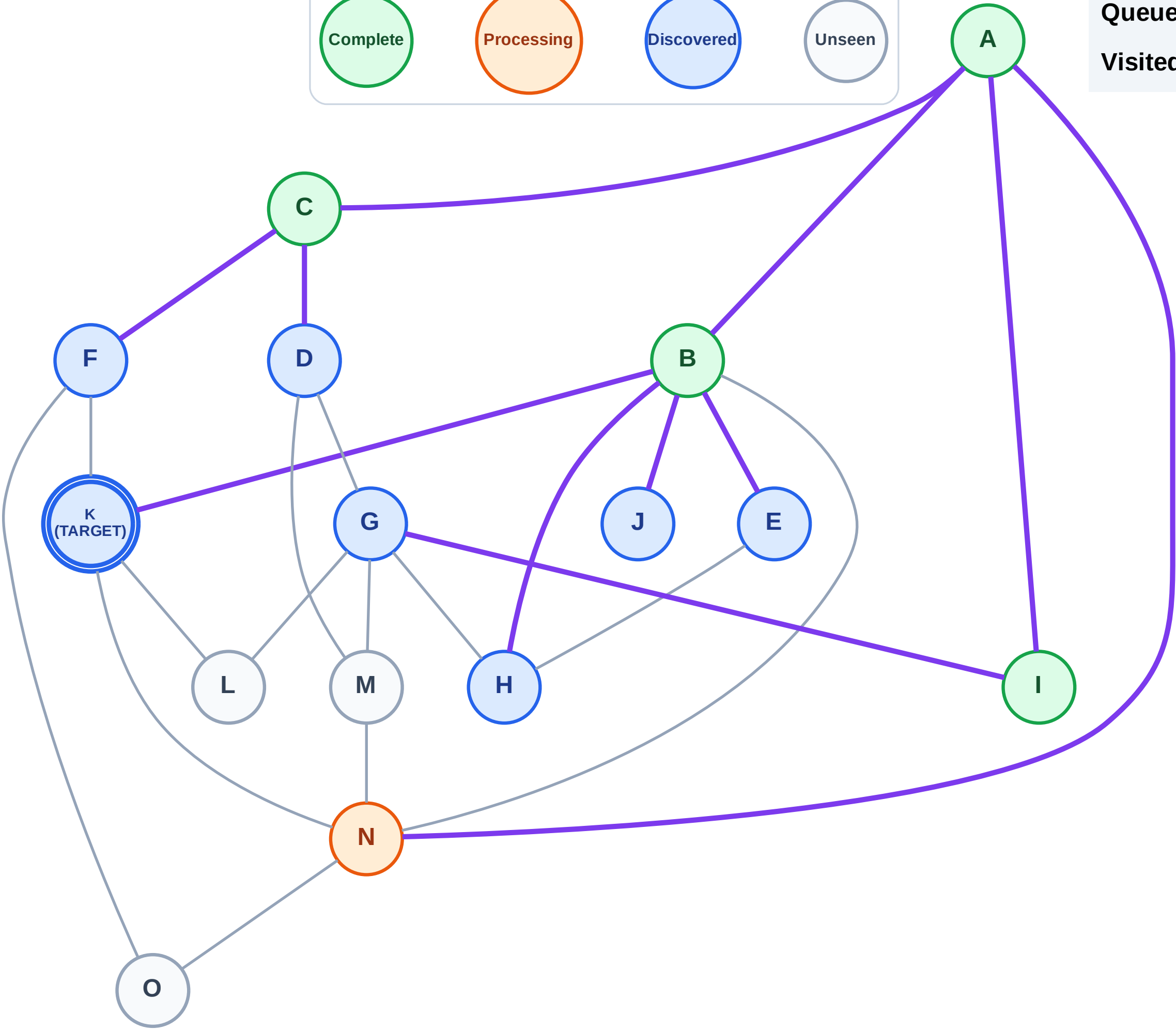
Processing

Discovered

Unseen

Queue: E → H → J → K → D → F → G

Visited: A, B, C, I



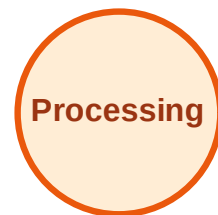
BFS Traversal

Step 11: Discover M, O from N

Legend



Complete



Processing



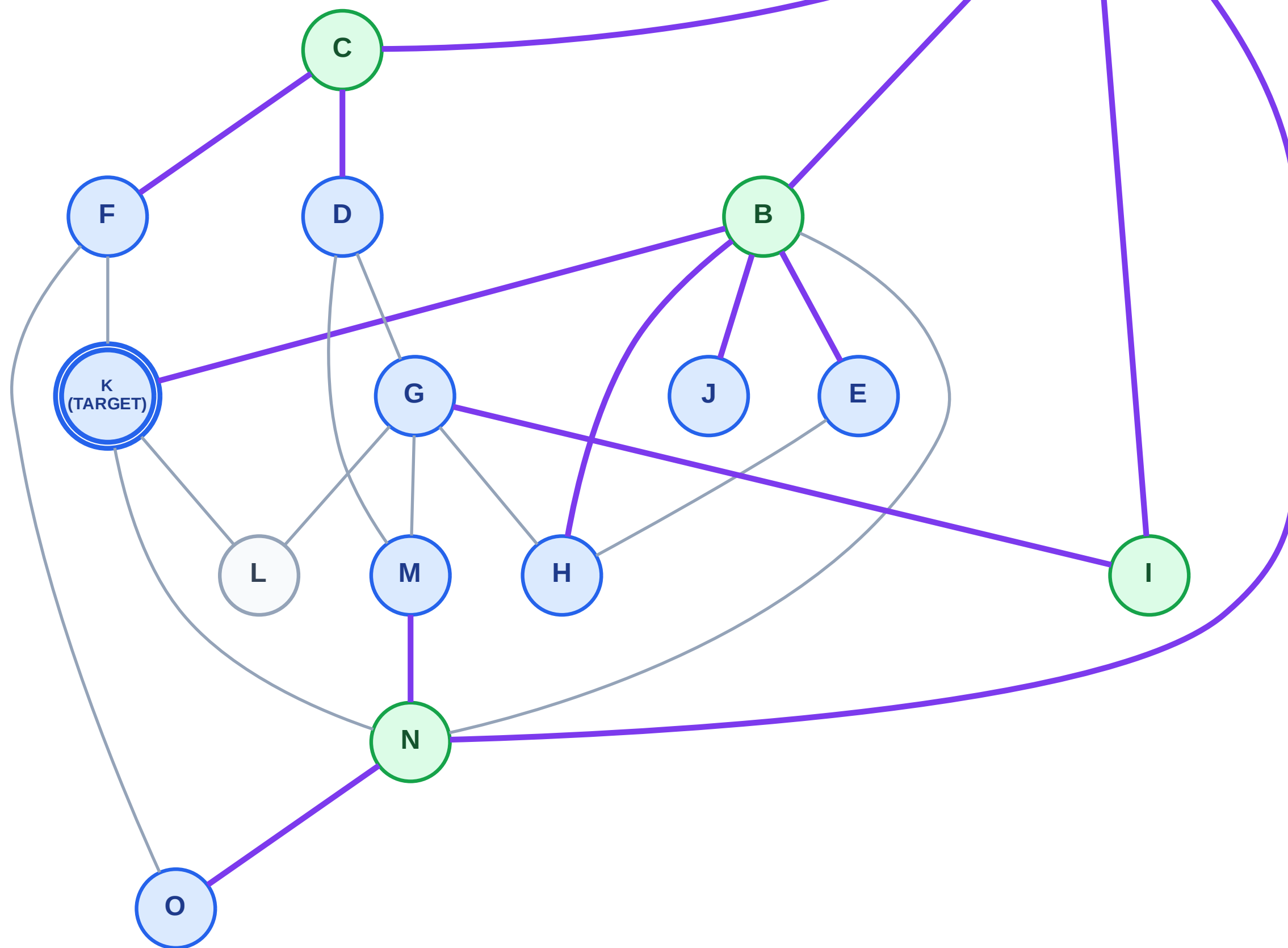
Discovered



Unseen

Queue: E → H → J → K → D → F → G → M → O

Visited: A, B, C, I, N



BFS Traversal

Step 12: Process node E

Legend



Complete



Processing



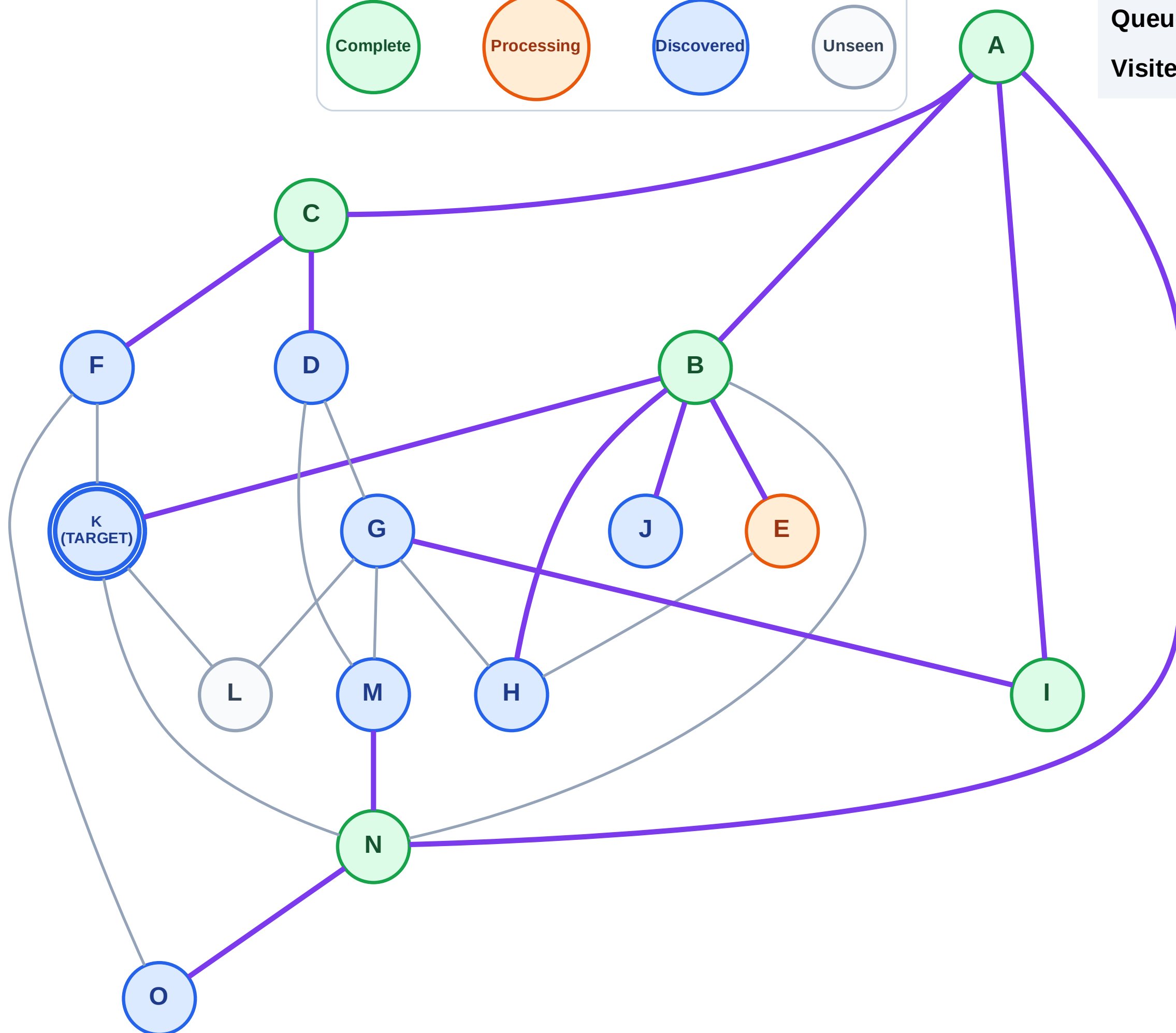
Discovered



Unseen

Queue: H → J → K → D → F → G → M → O

Visited: A, B, C, I, N



BFS Traversal

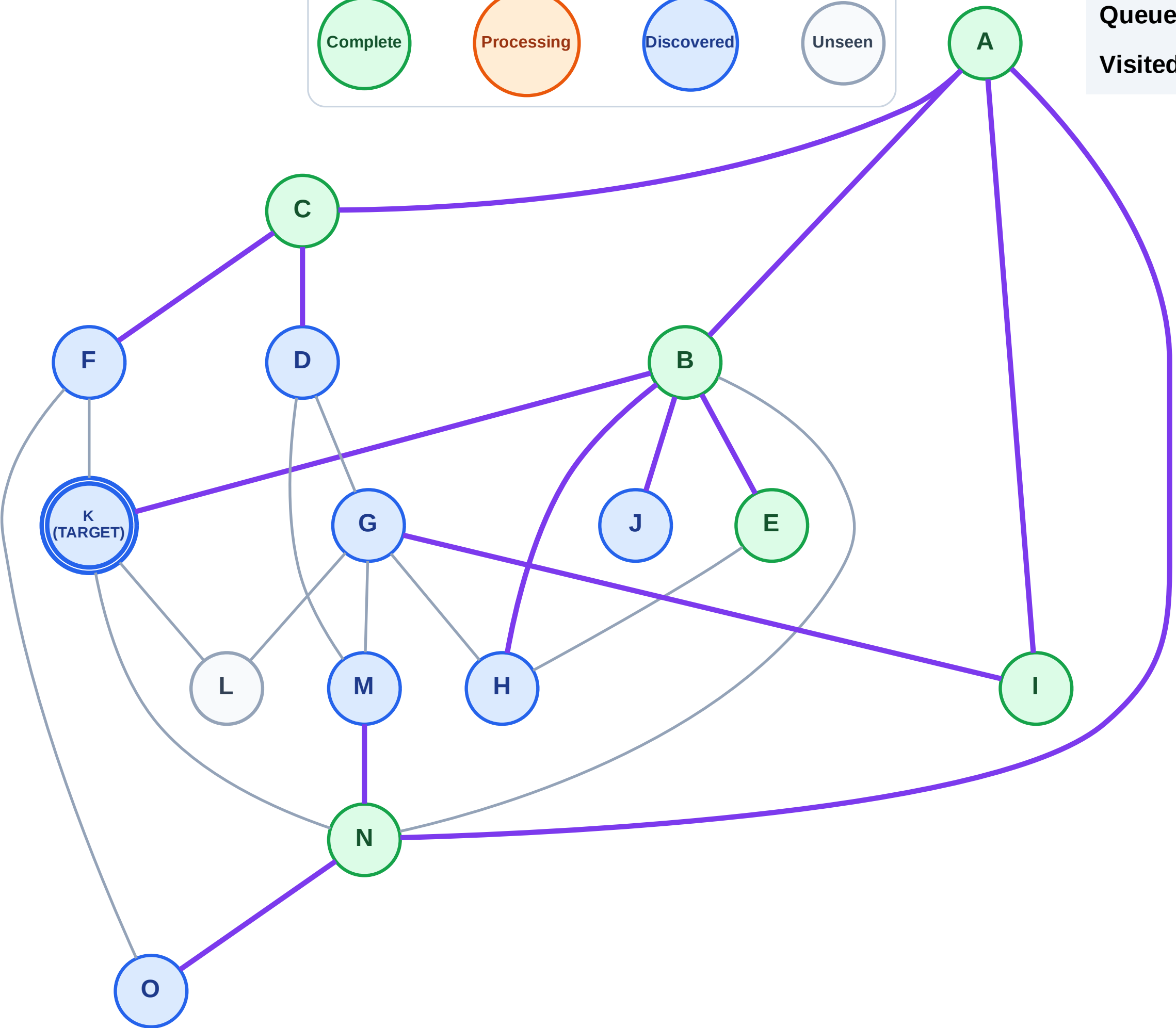
Step 13: No new neighbors from E

Legend



Queue: H → J → K → D → F → G → M → O

Visited: A, B, C, E, I, N



BFS Traversal

Step 14: Process node H

Legend

Complete

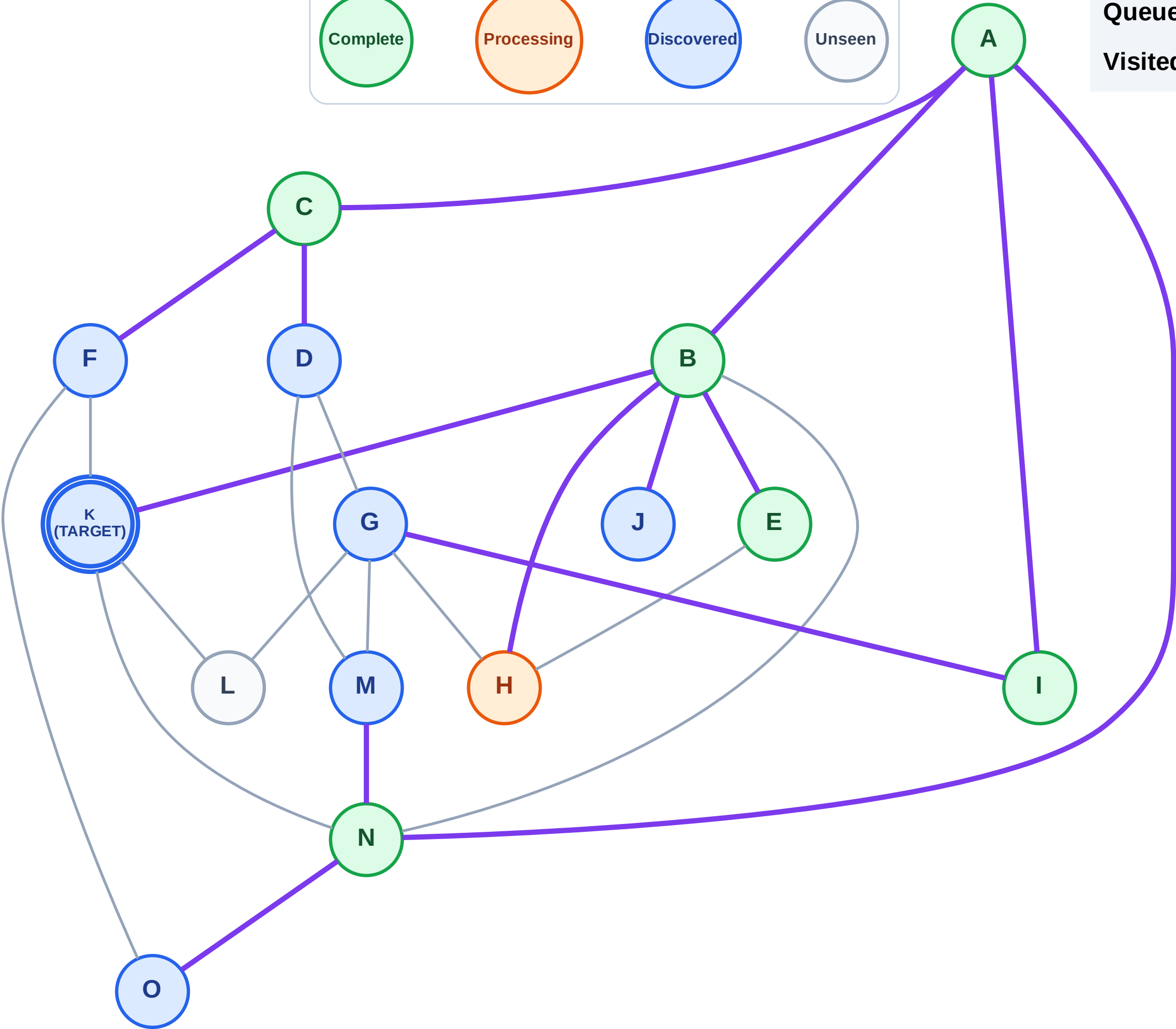
Processing

Discovered

Unseen

Queue: J → K → D → F → G → M → O

Visited: A, B, C, E, I, N



BFS Traversal

Step 15: No new neighbors from H

Legend

Complete

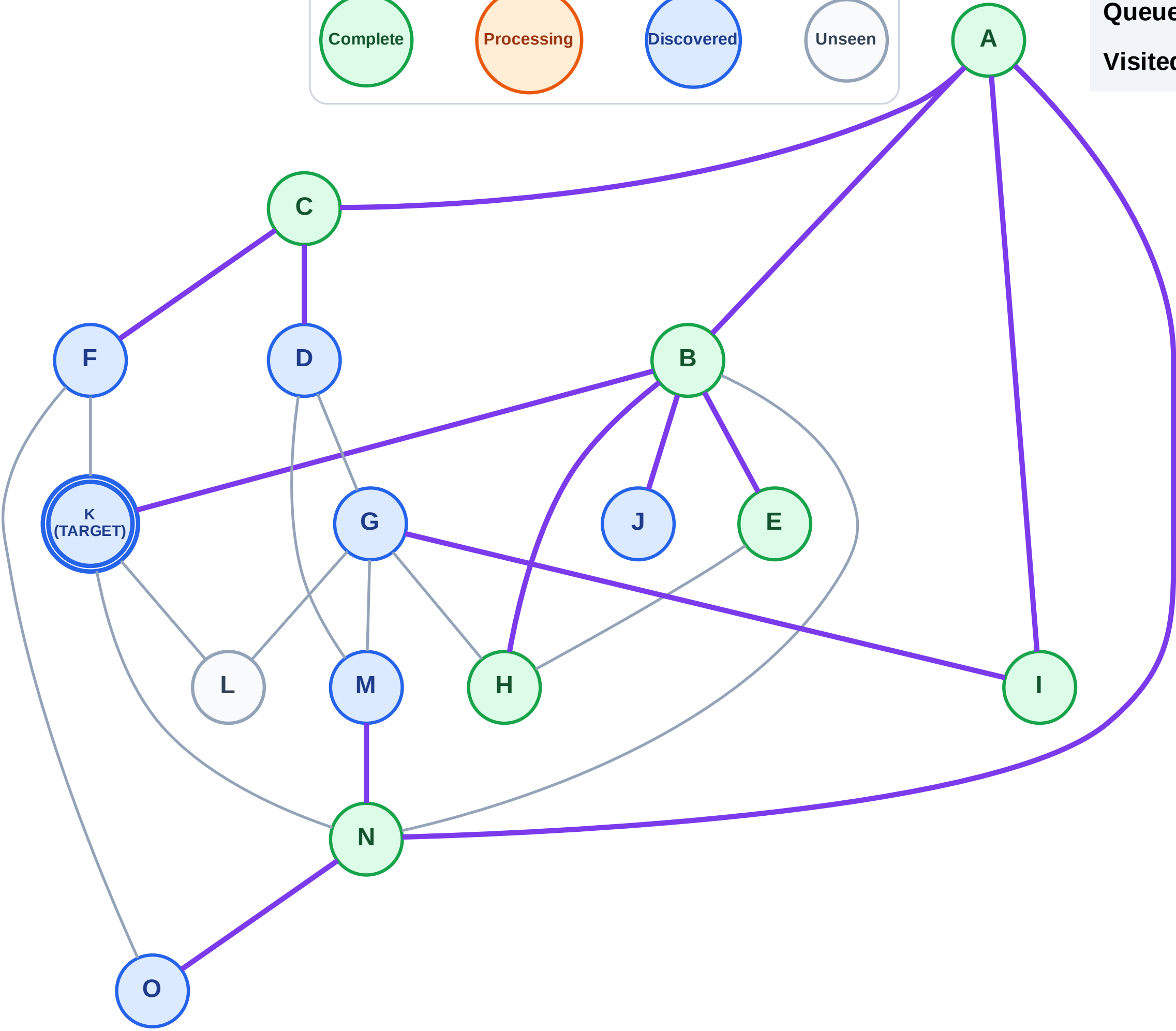
Processing

Discovered

Unseen

Queue: J → K → D → F → G → M → O

Visited: A, B, C, E, H, I, N



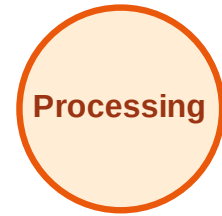
BFS Traversal

Step 16: Process node J

Legend



Complete



Processing



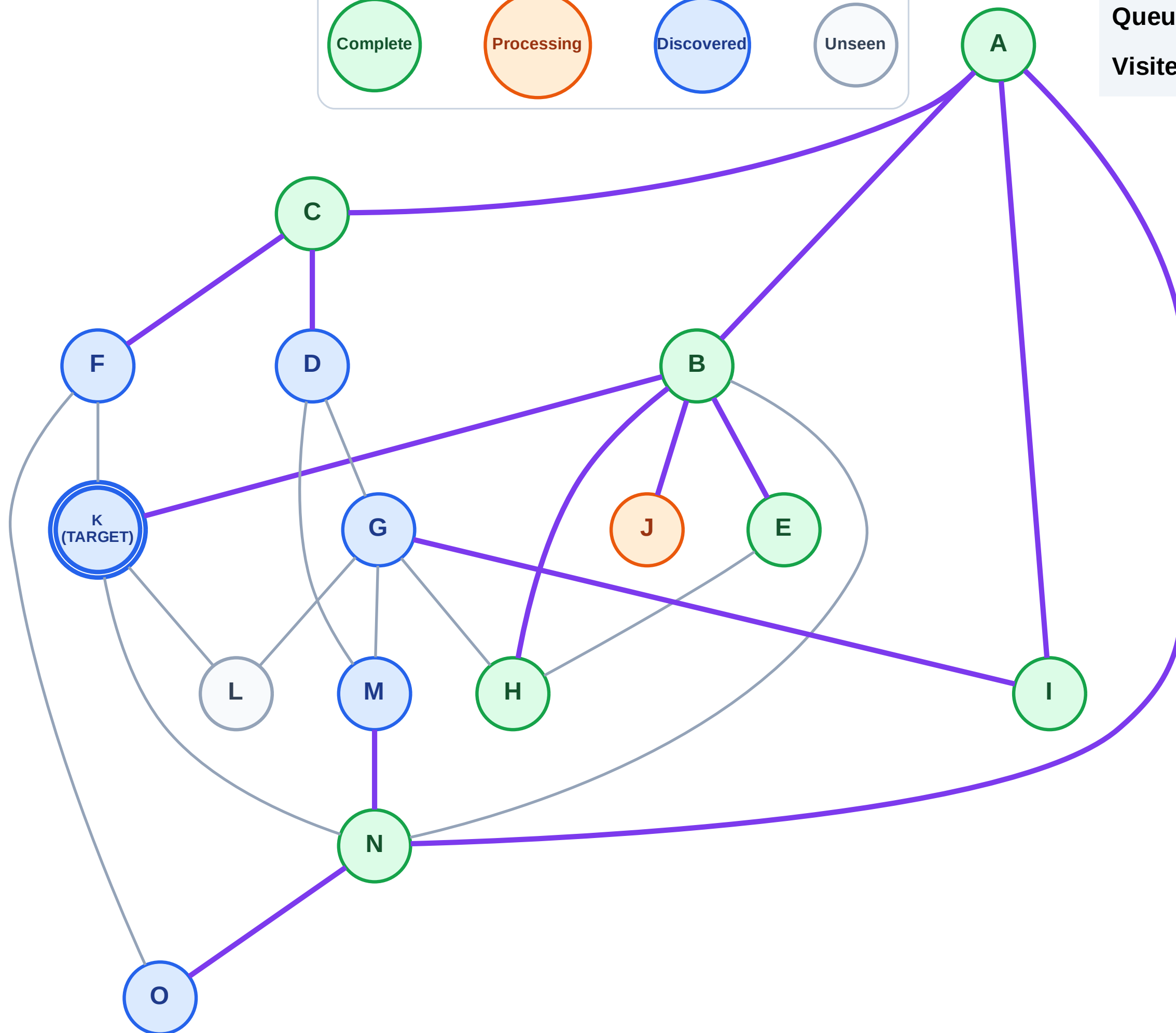
Discovered



Unseen

Queue: K → D → F → G → M → O

Visited: A, B, C, E, H, I, N



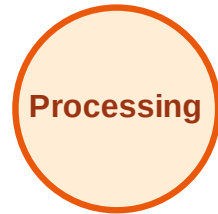
BFS Traversal

Step 17: No new neighbors from J

Legend



Complete



Processing



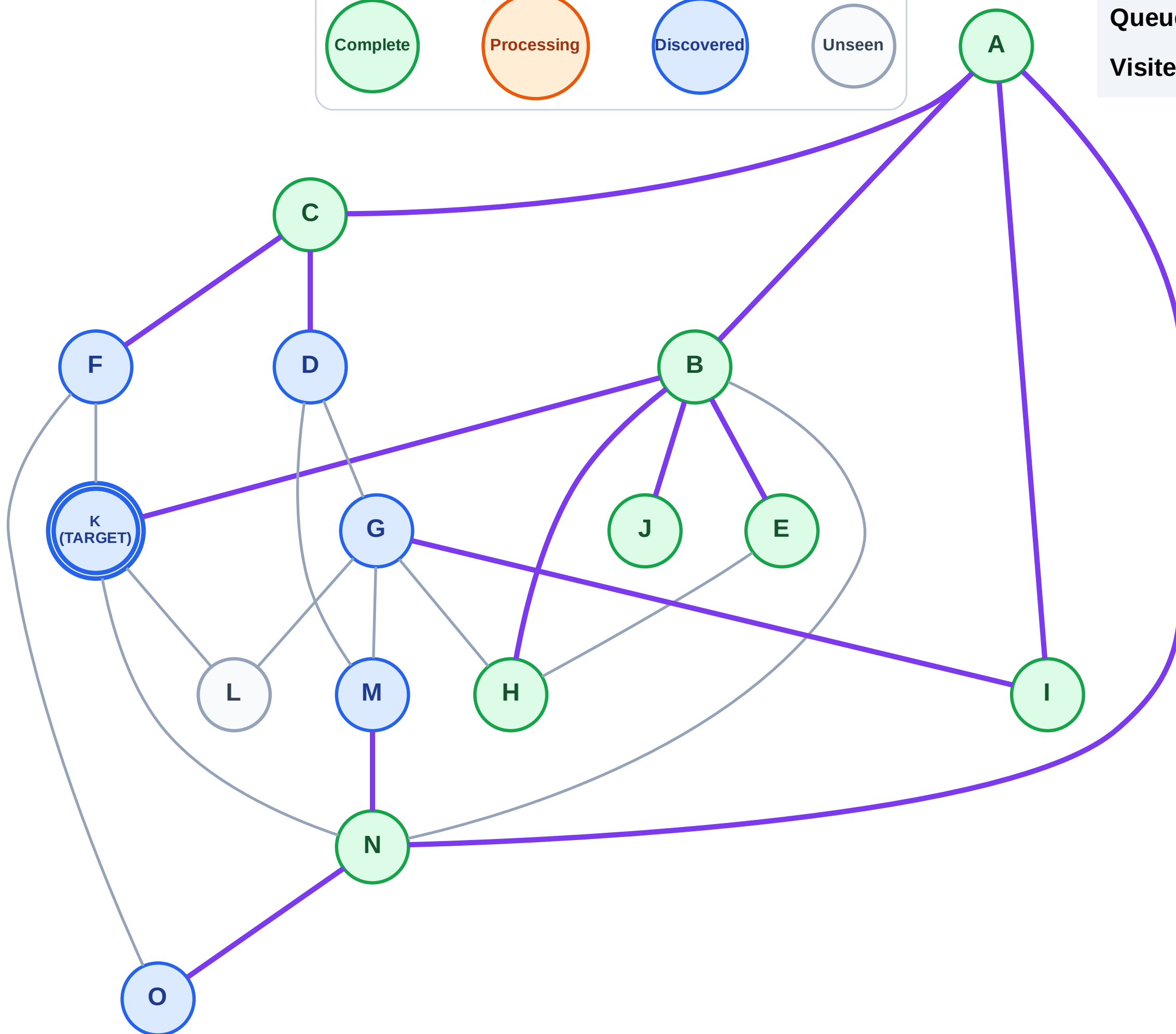
Discovered



Unseen

Queue: K → D → F → G → M → O

Visited: A, B, C, E, H, I, J, N



BFS Traversal

Step 18: Process node K

Legend

Complete

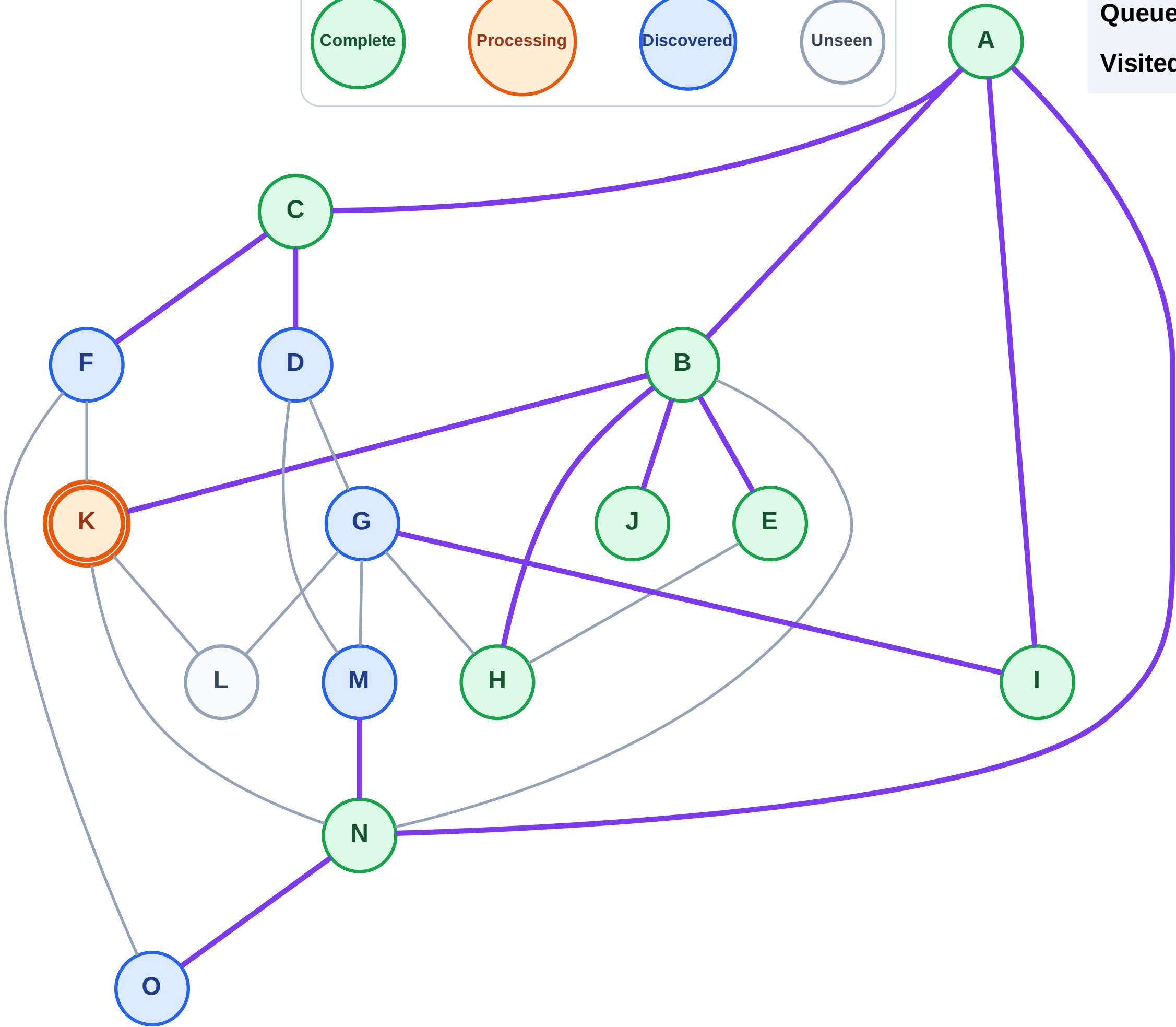
Processing

Discovered

Unseen

Queue: D → F → G → M → O

Visited: A, B, C, E, H, I, J, N



BFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

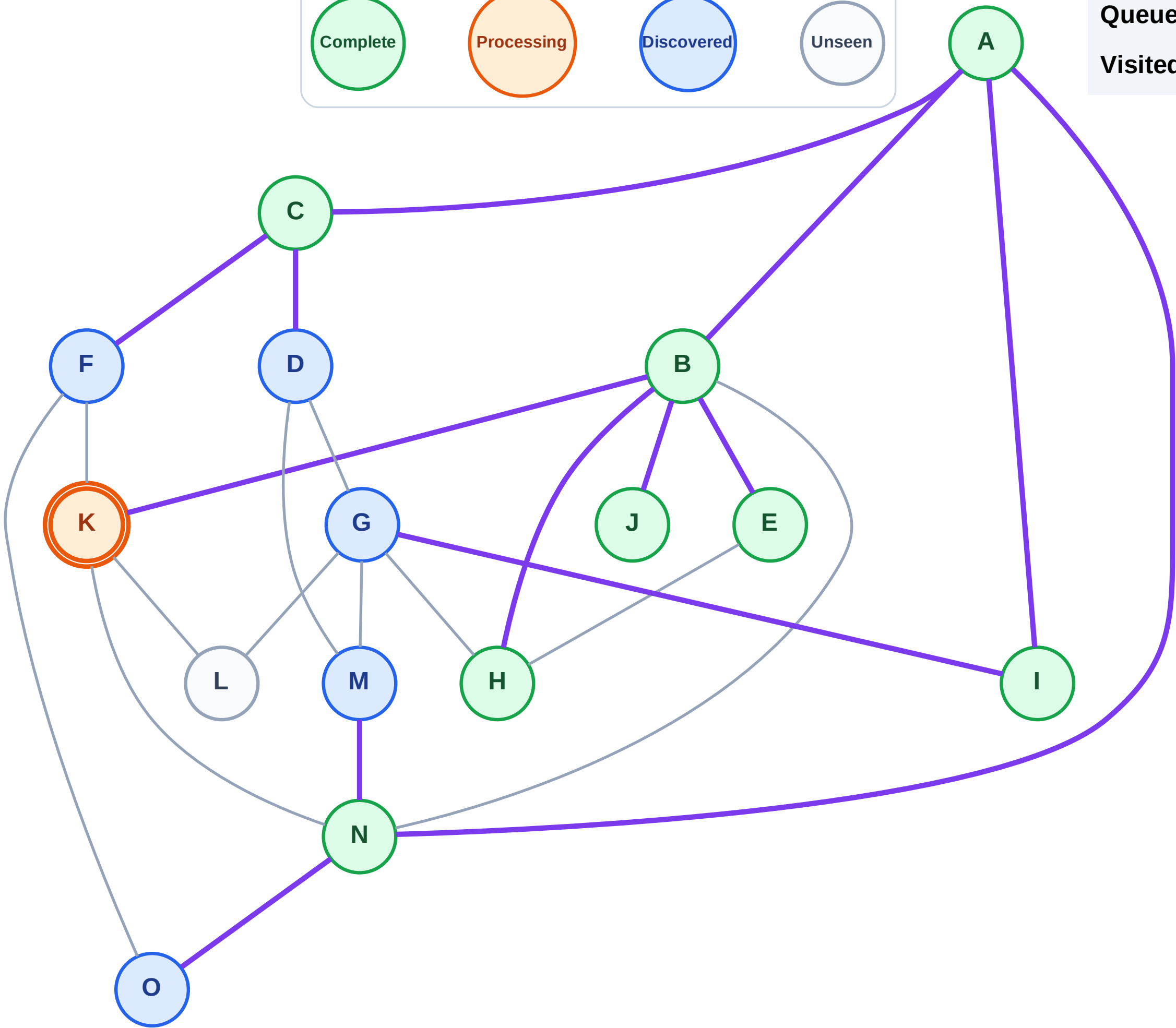
Processing

Discovered

Unseen

Queue: D → F → G → M → O

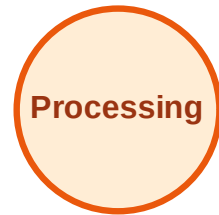
Visited: A, B, C, E, H, I, J, K, N



DFS Traversal

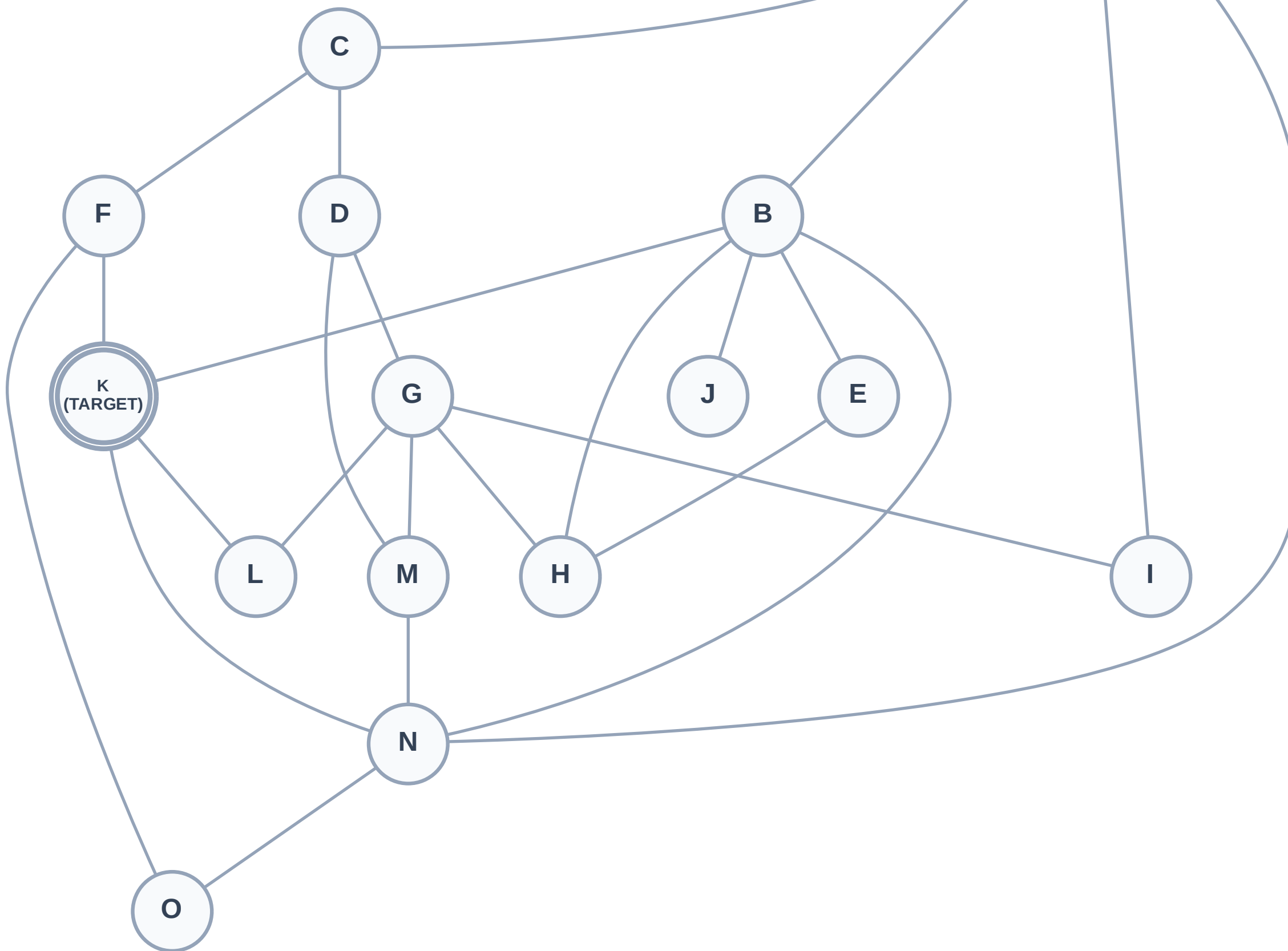
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

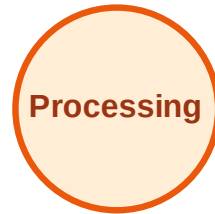
Visited: (none)



DFS Traversal

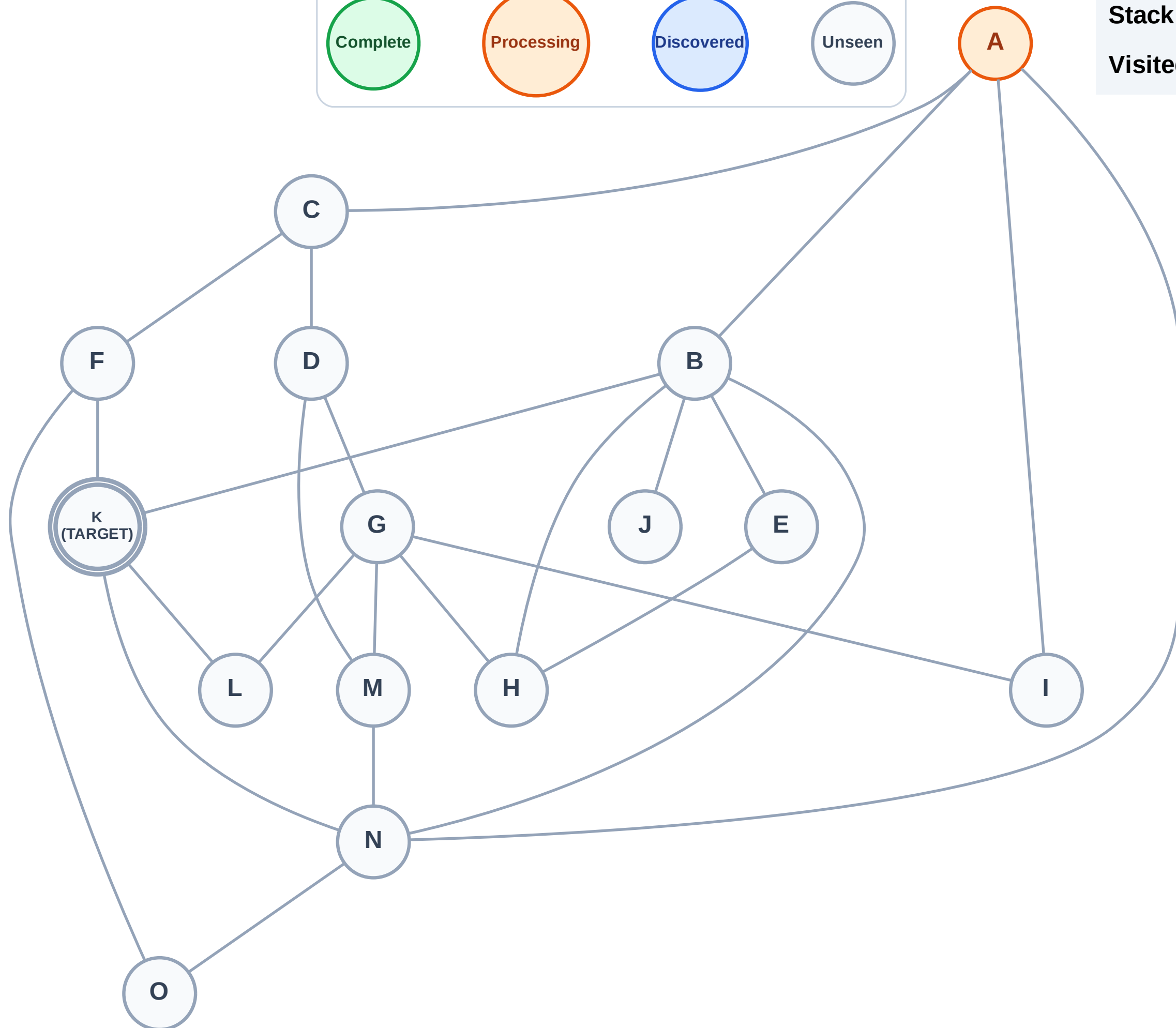
Step 2: Process node A

Legend



Stack (top at right): (empty)

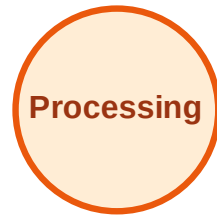
Visited: (none)



DFS Traversal

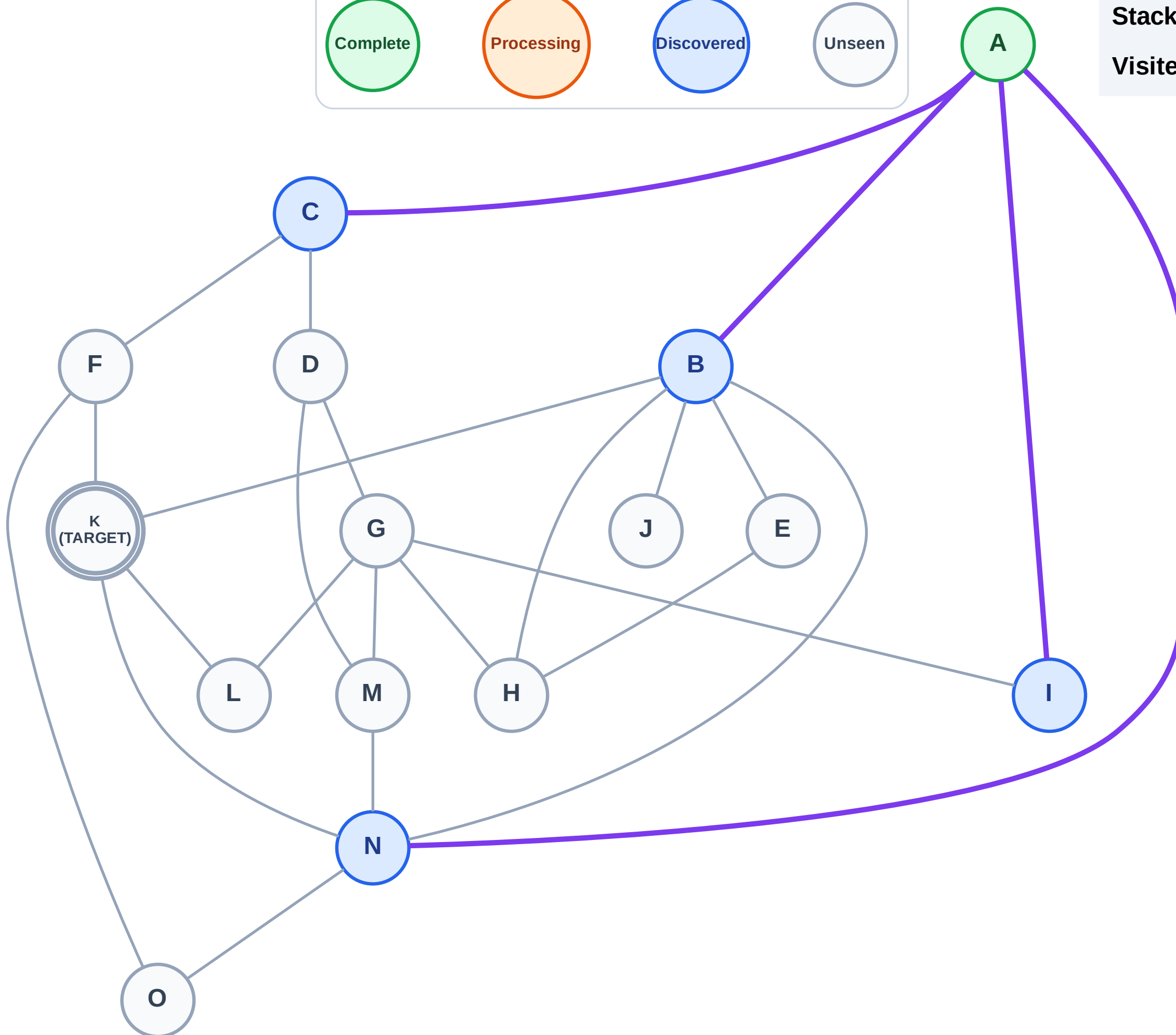
Step 3: Discover N, I, C, B from A

Legend



Stack (top at right): N | I | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

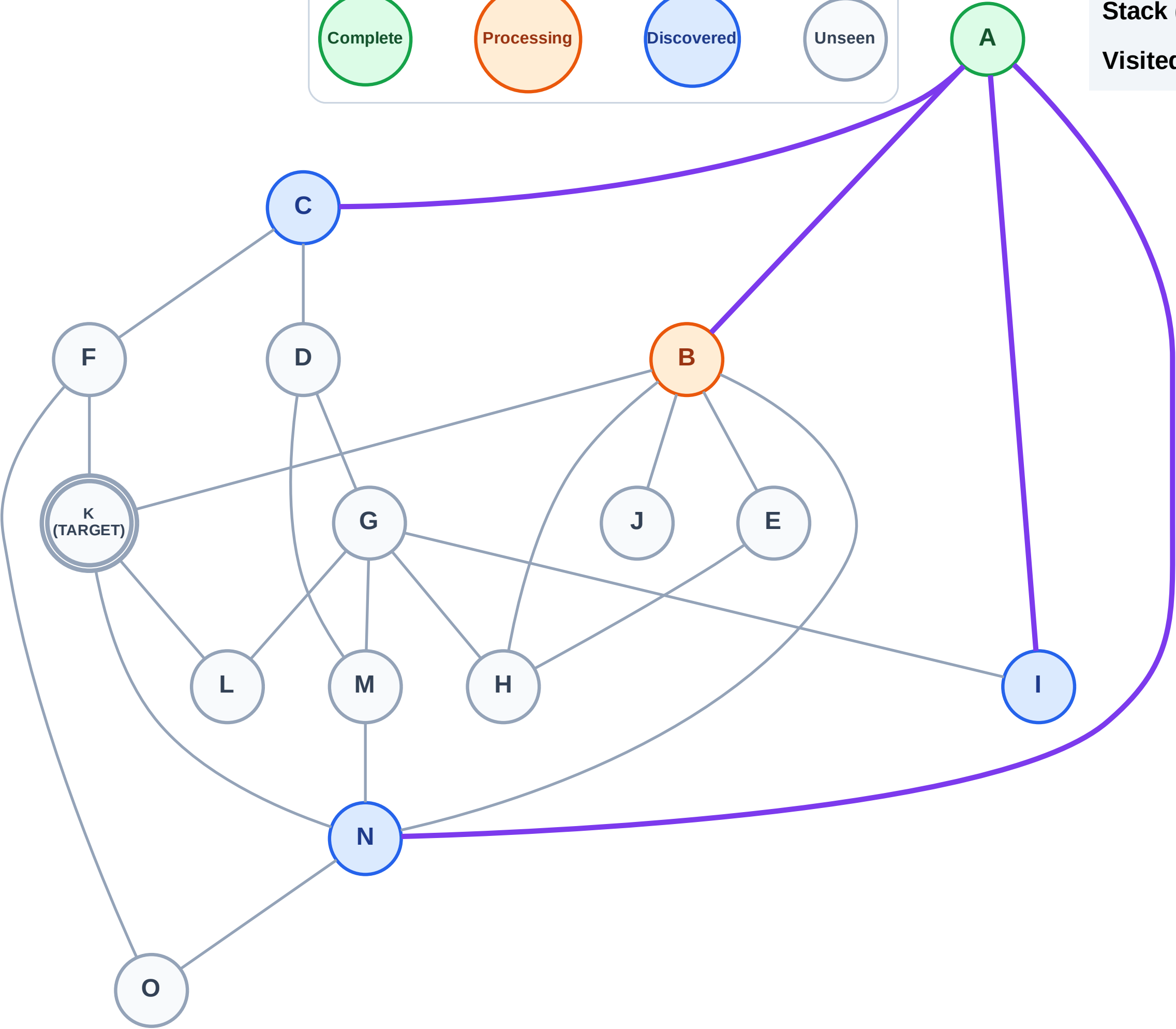
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C
Visited: A



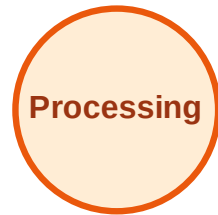
DFS Traversal

Step 5: Discover K, J, H, E from B

Legend



Complete



Processing



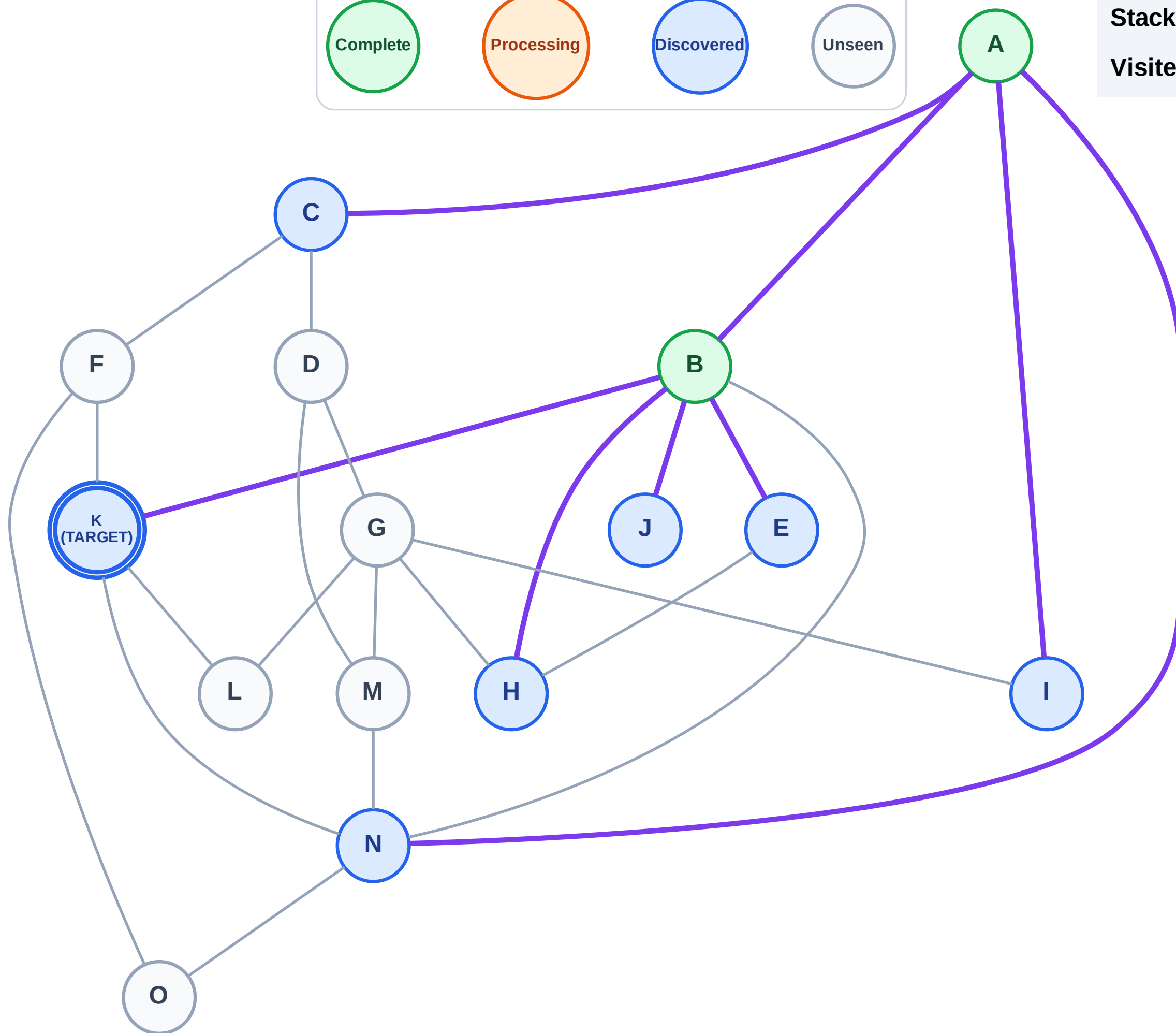
Discovered



Unseen

Stack (top at right): N | I | C | K | J | H | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

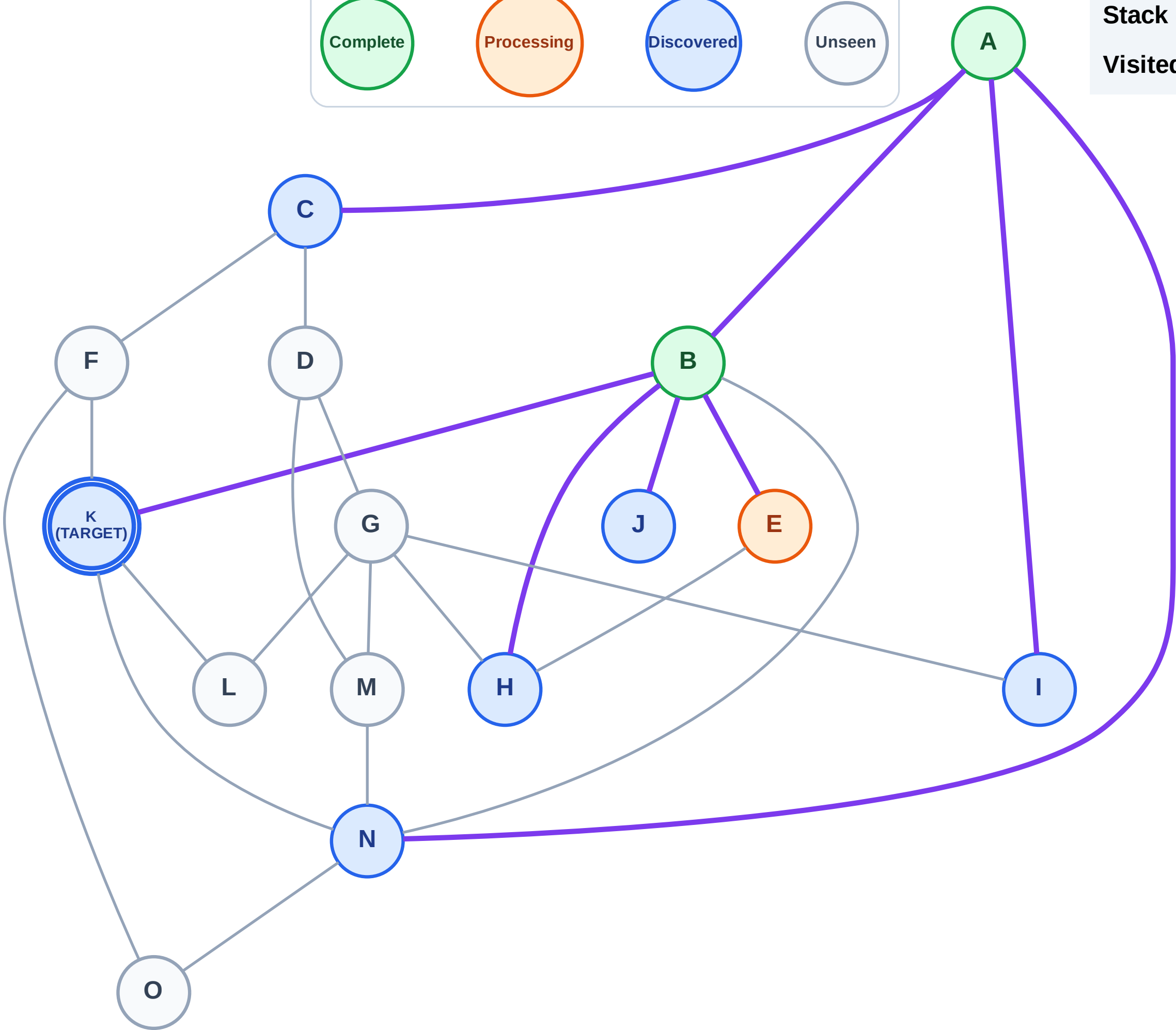
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J | H

Visited: A, B



Step 7: No new neighbors from E

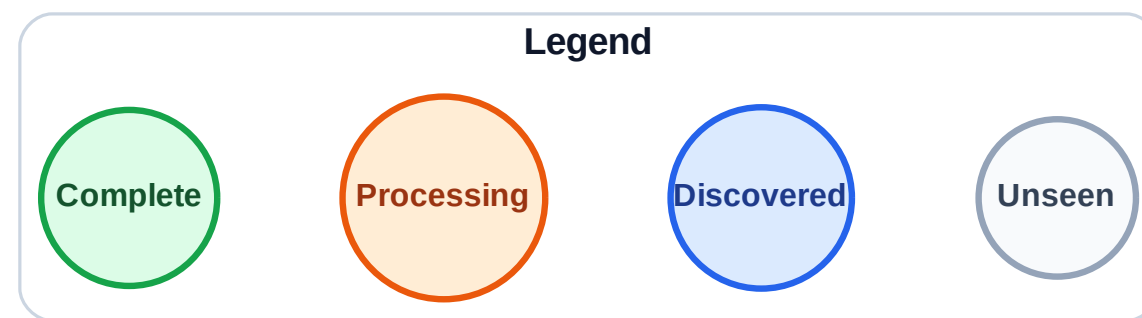
Legend



The graph shows a search process with nodes A through O. Nodes are colored based on their status: Complete (green), Processing (orange), Discovered (blue), and Unseen (grey). The graph shows a path from A to B to E, with other nodes like C, D, F, G, H, I, J, K, L, M, N, and O connected by edges. A stack on the right shows the current state: N | I | C | K | J | H. Visited nodes are A, B, and E.

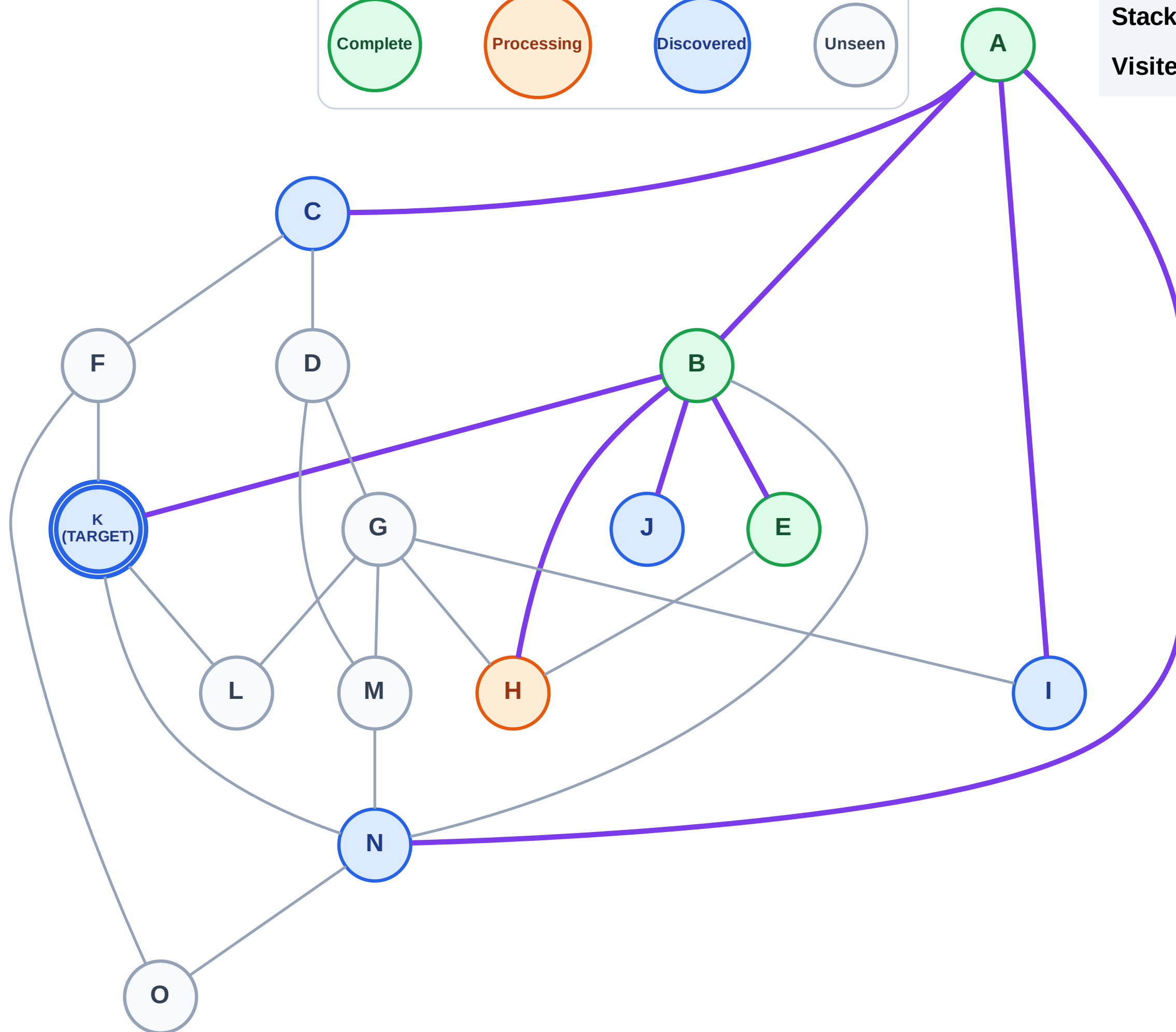
Step 8: Process node H

Step 8: Process node H



Stack (top at right): N | I | C | K | J

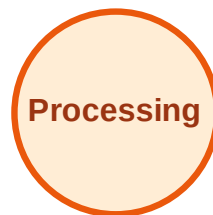
Visited: A, B, E



DFS Traversal

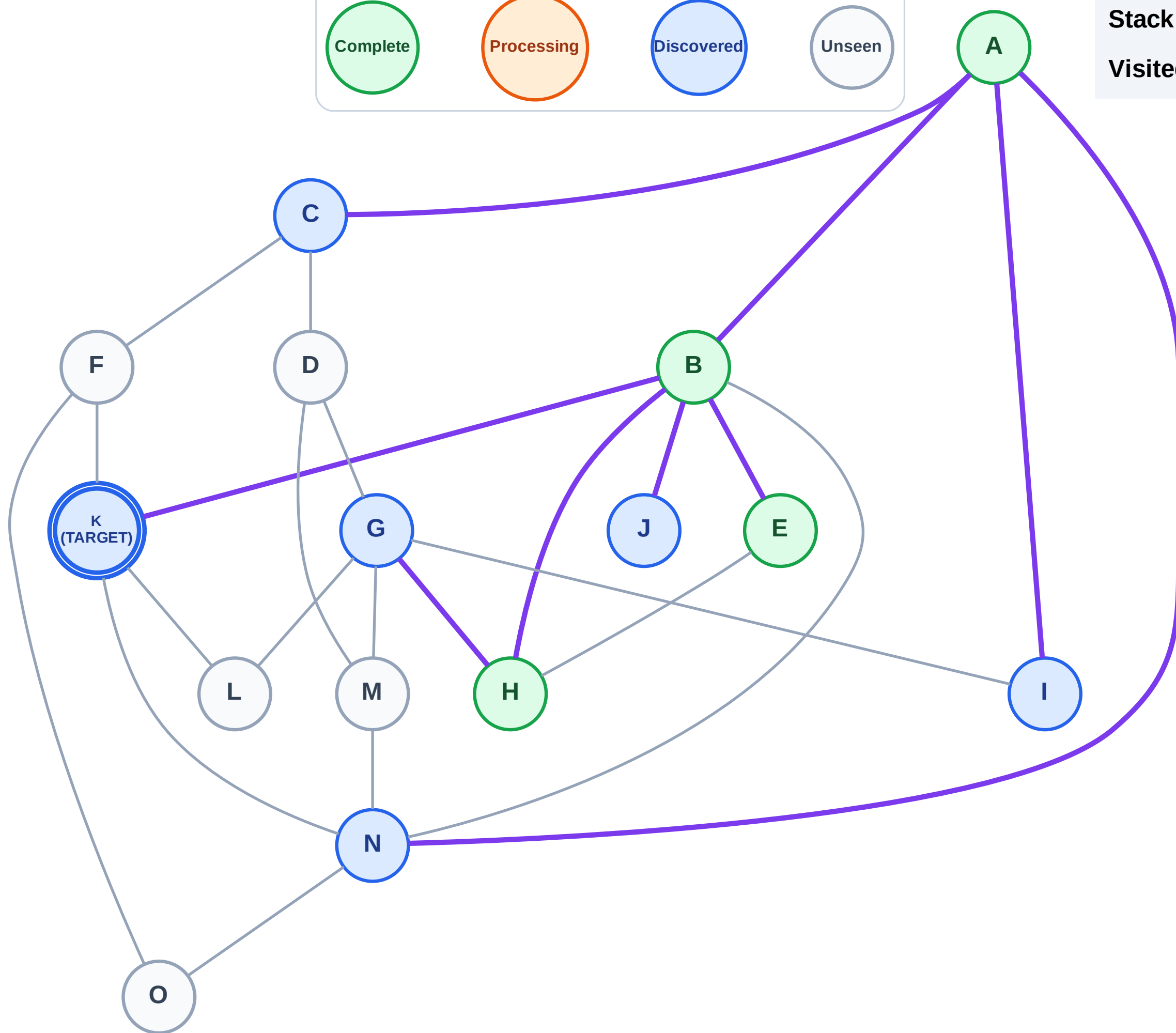
Step 9: Discover G from H

Legend



Stack (top at right): N | I | C | K | J | G

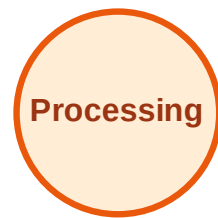
Visited: A, B, E, H



DFS Traversal

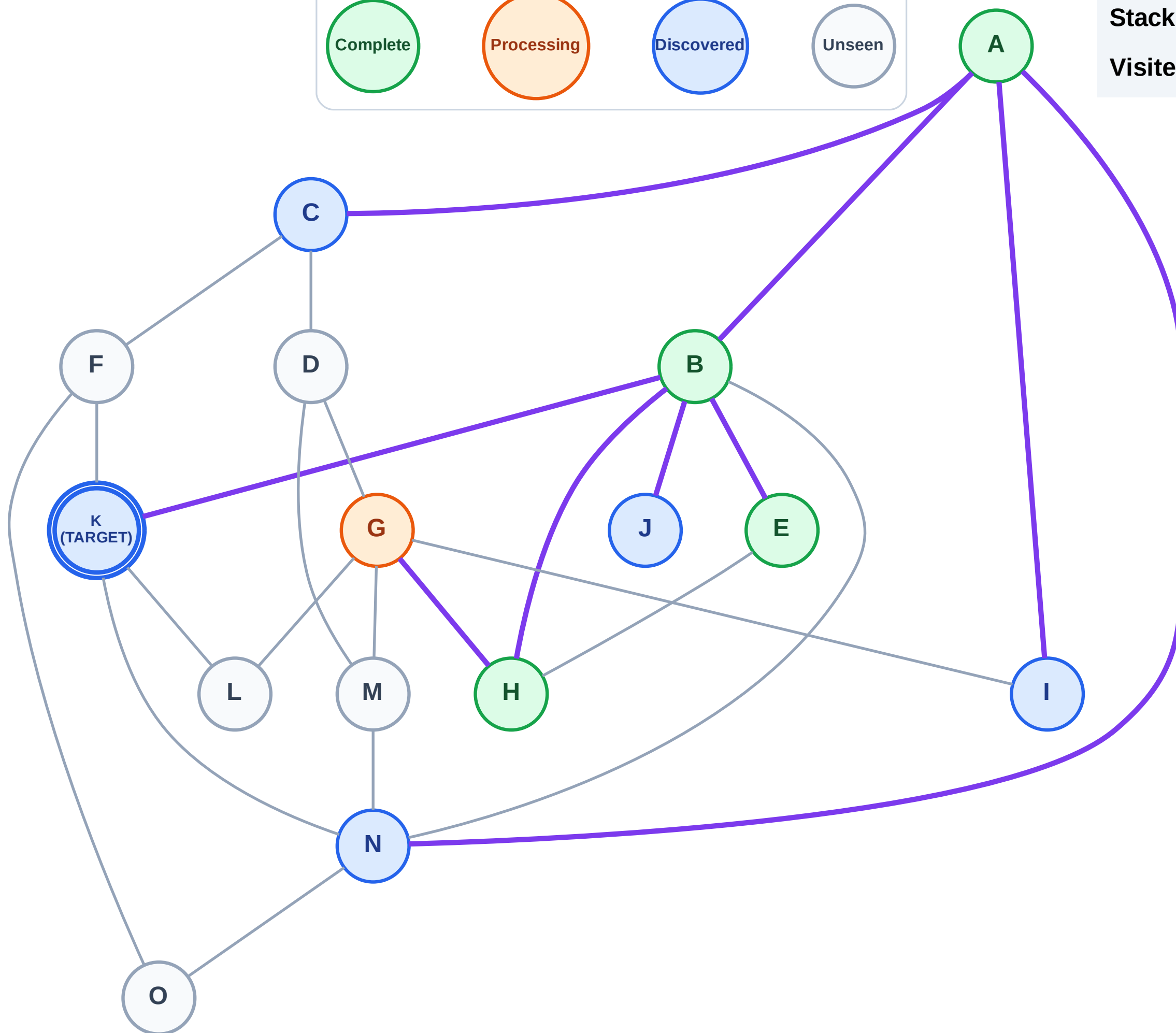
Step 10: Process node G

Legend



Stack (top at right): N | I | C | K | J

Visited: A, B, E, H



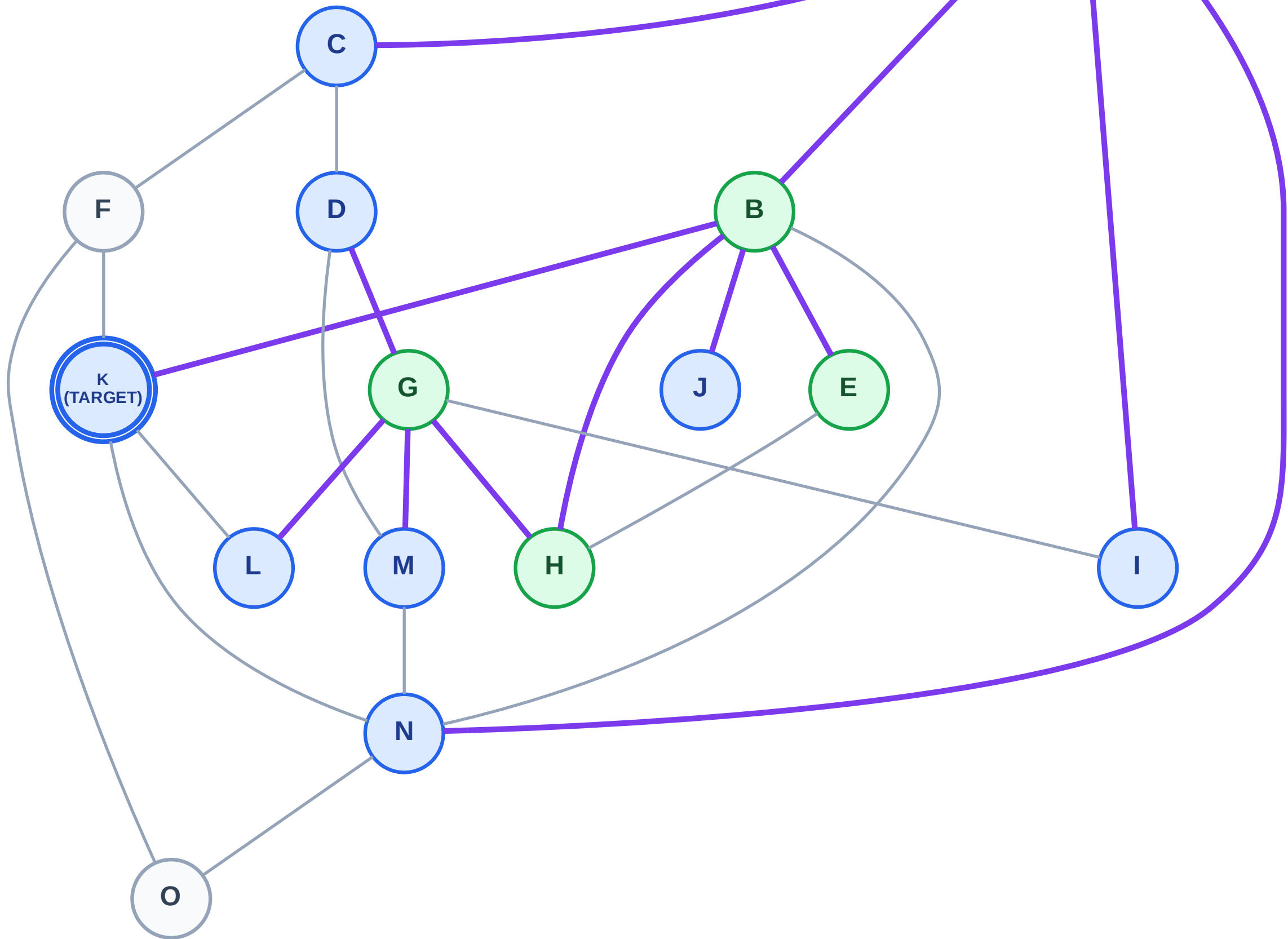
Step 11: Discover M, L, D from G

Stack (top at right): N | I | C | K | J | M | L | D

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.



DFS Traversal

Step 12: Process node D

Legend

Complete

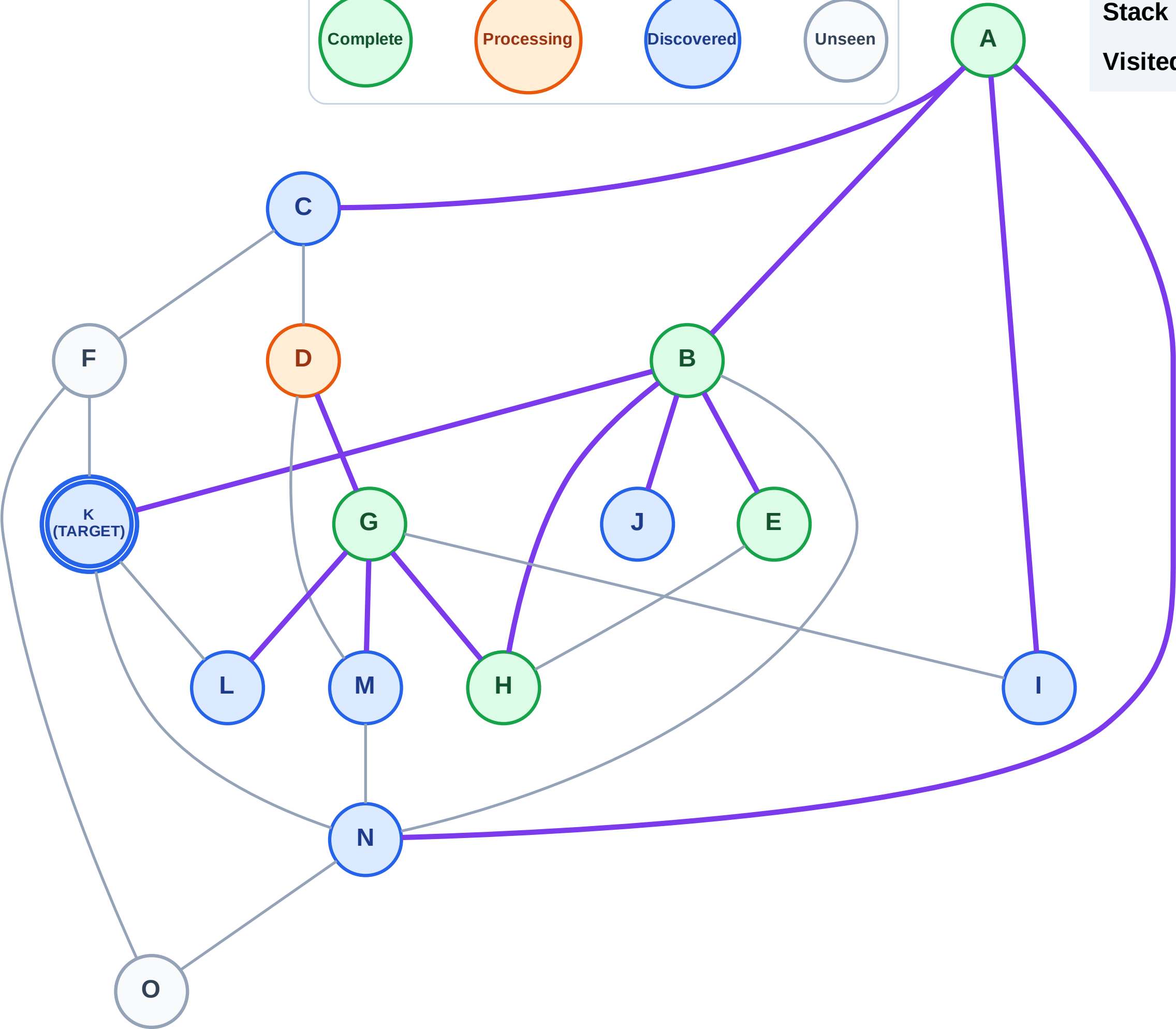
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J | M | L

Visited: A, B, E, G, H



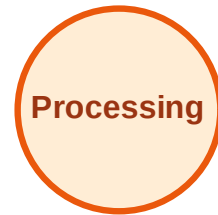
DFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing



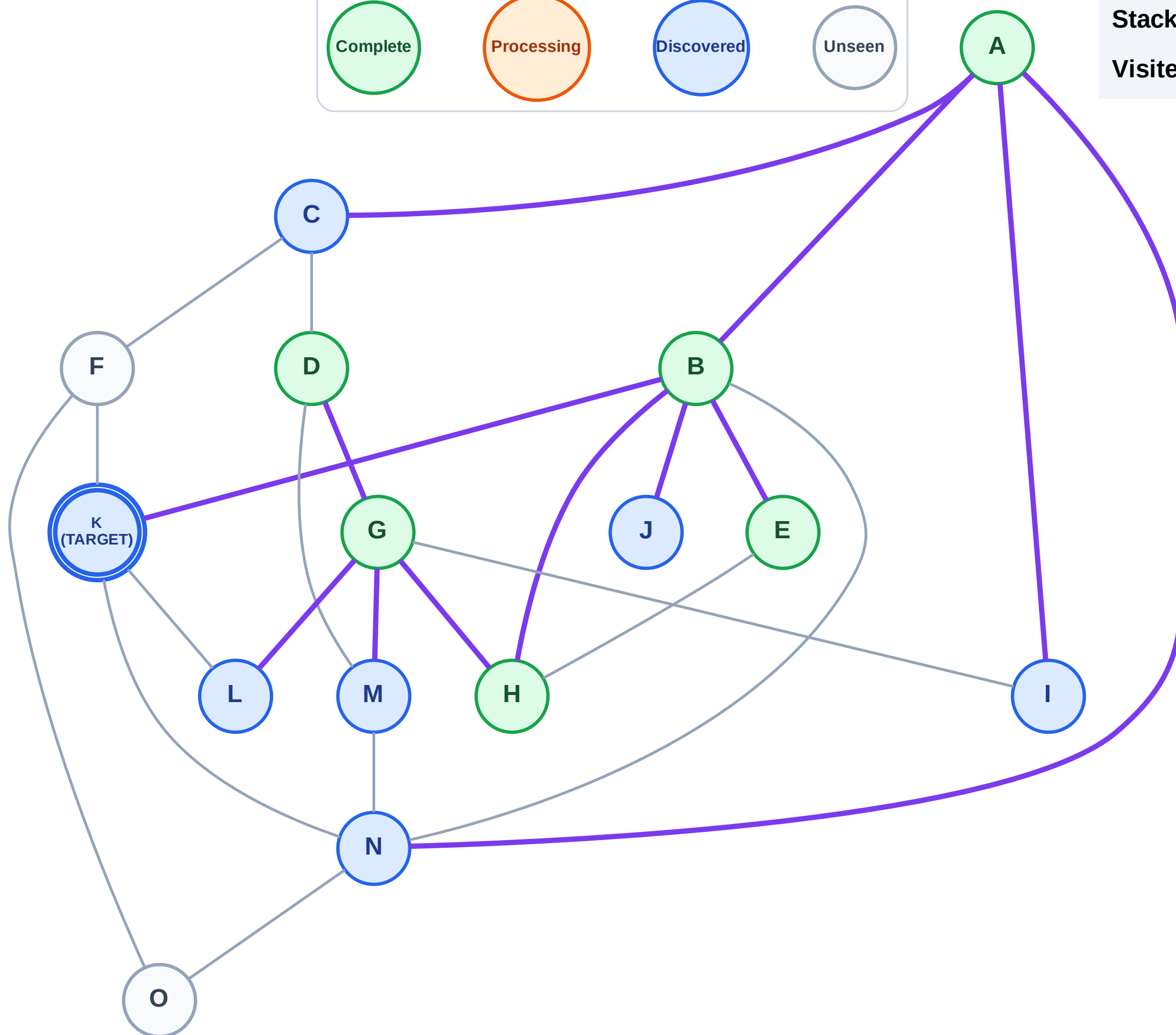
Discovered



Unseen

Stack (top at right): N | I | C | K | J | M | L

Visited: A, B, D, E, G, H



DFS Traversal

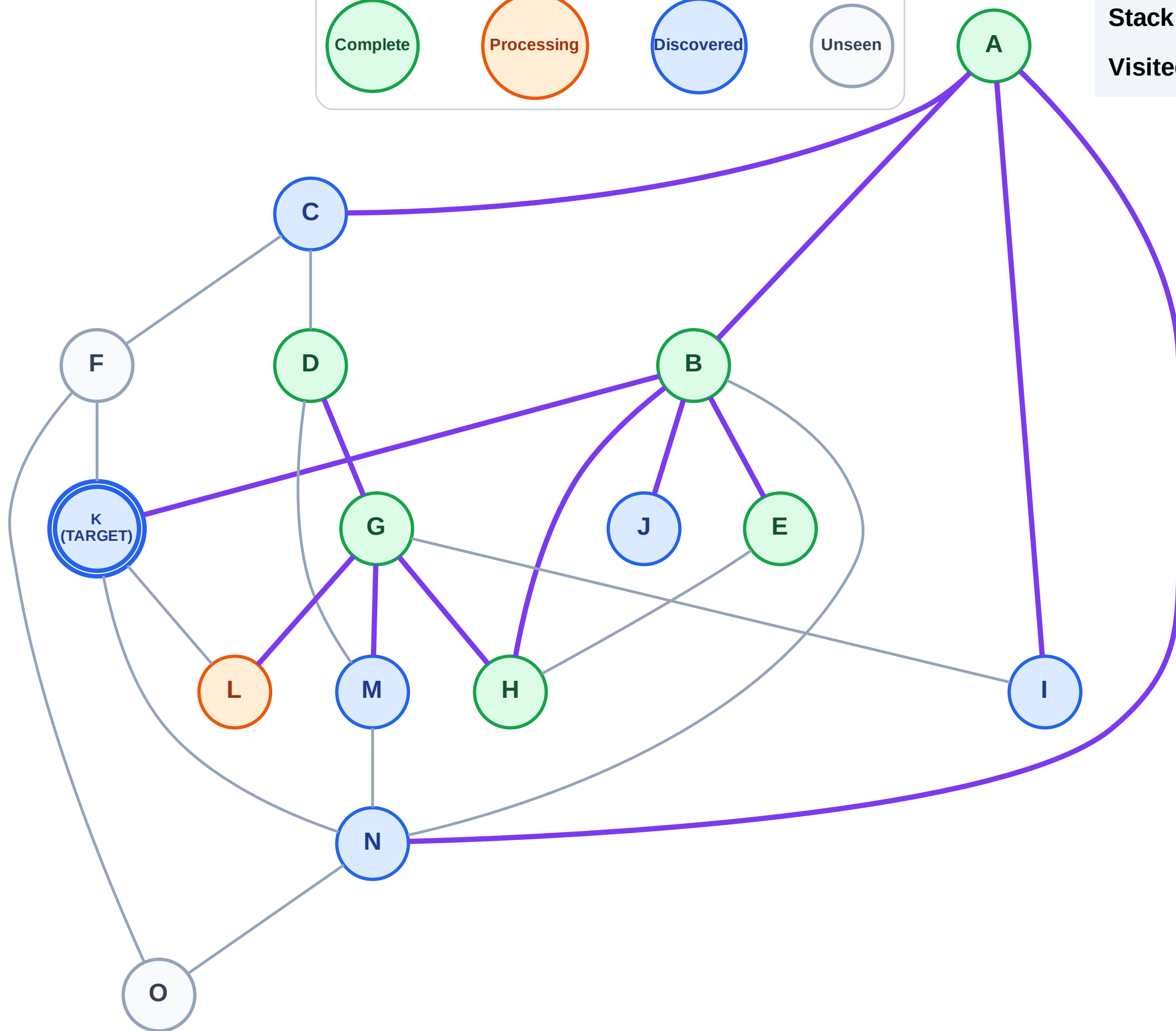
Step 14: Process node L

Legend



Stack (top at right): N | I | C | K | J | M

Visited: A, B, D, E, G, H

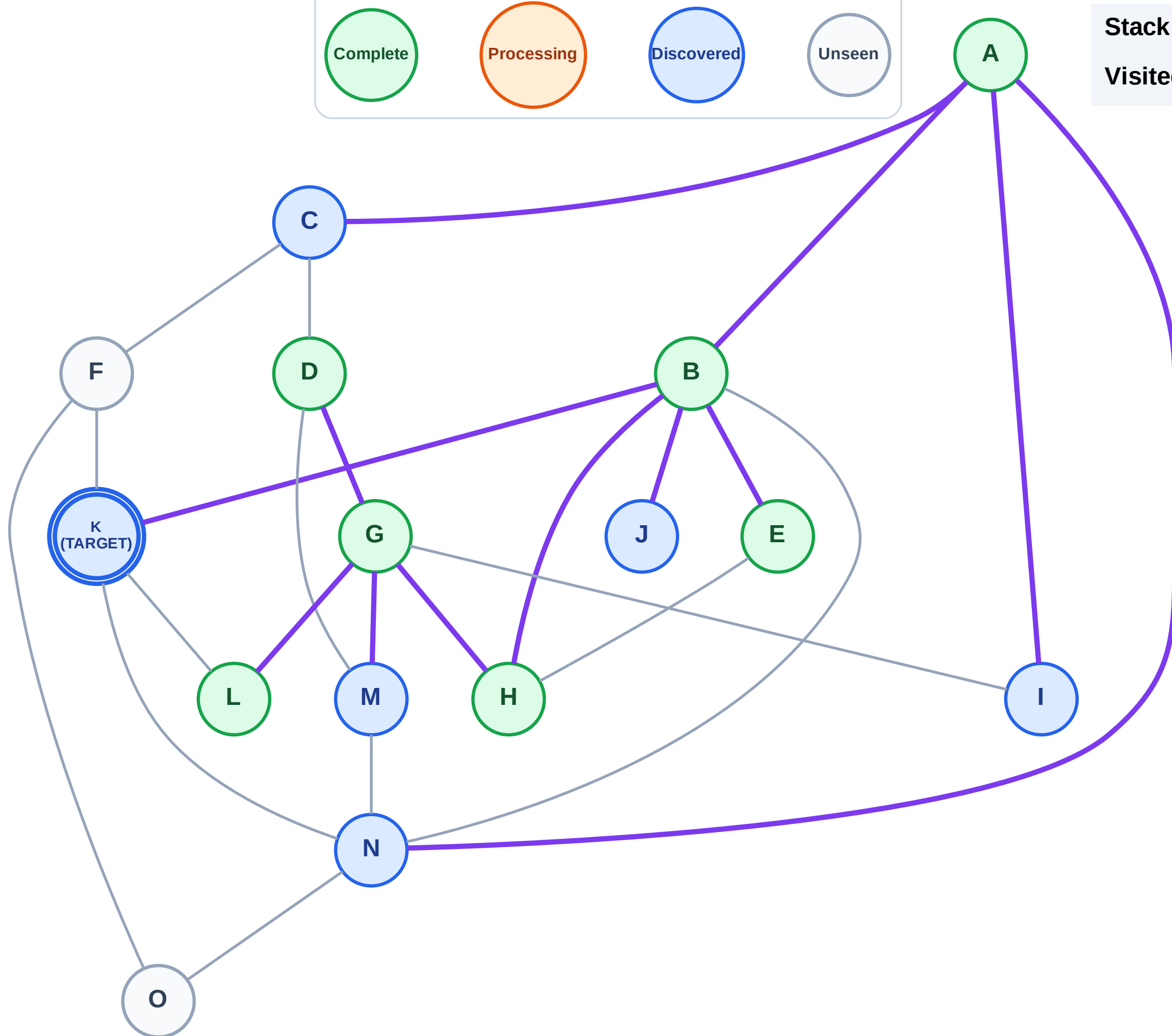


Step 15: No new neighbors from L

Legend

Unseen

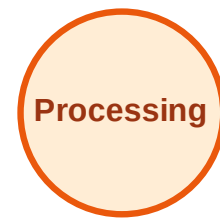
Visited: A, B, D, E, G, H, L



DFS Traversal

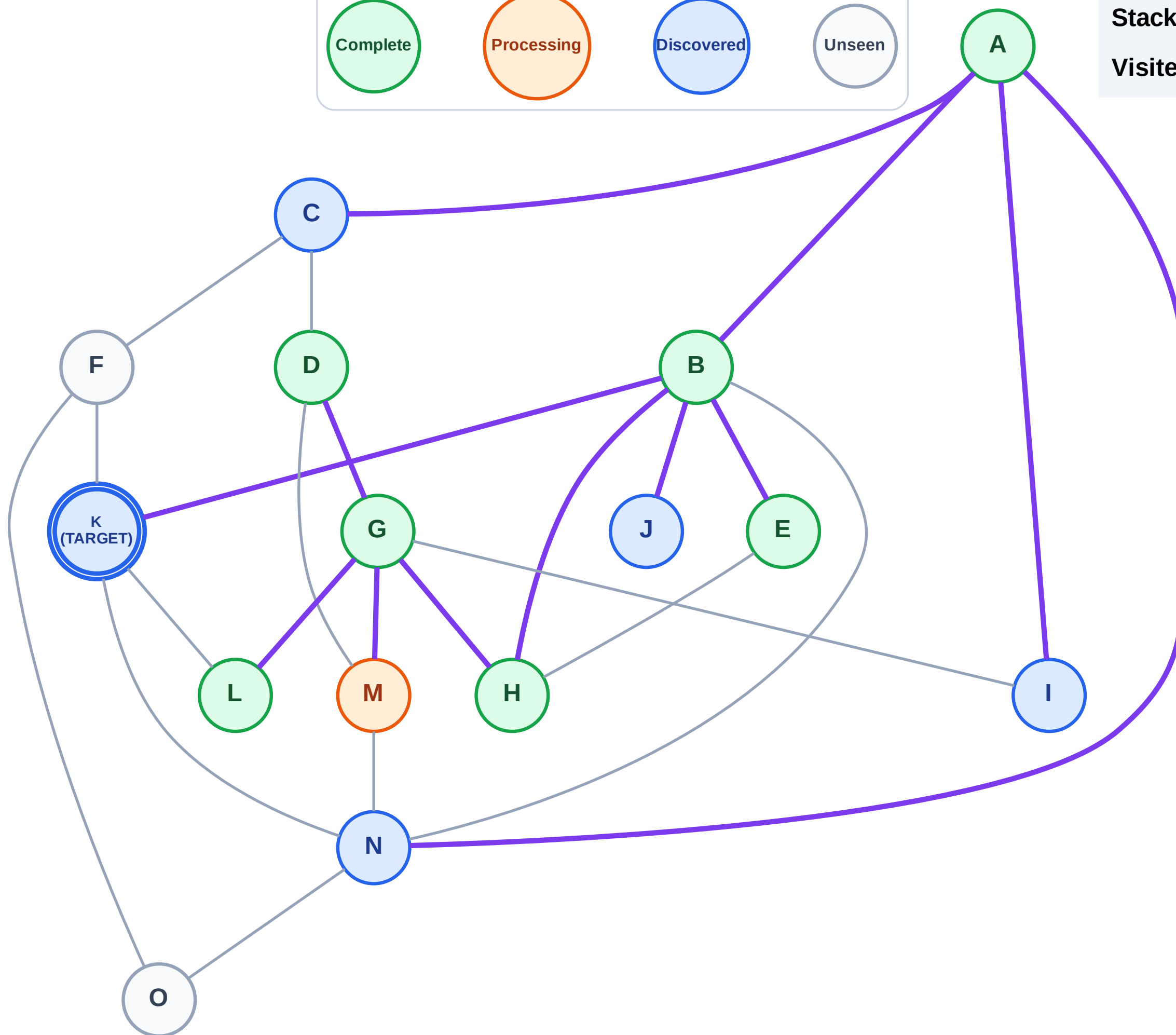
Step 16: Process node M

Legend



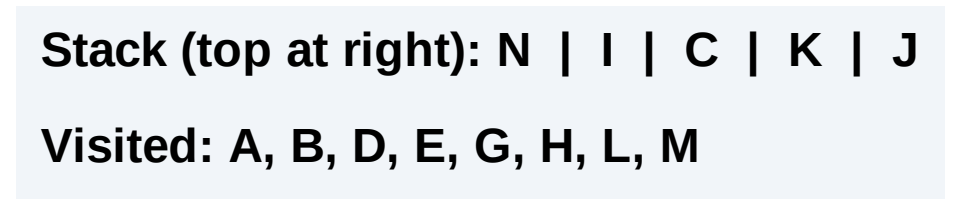
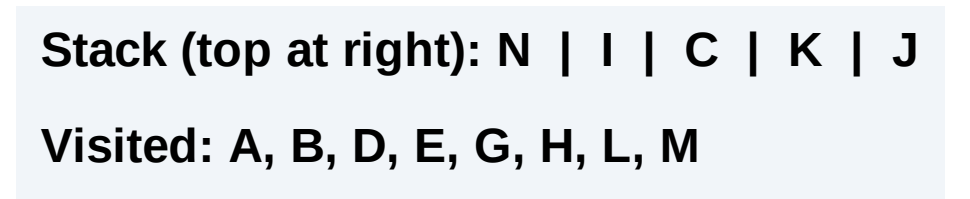
Stack (top at right): N | I | C | K | J

Visited: A, B, D, E, G, H, L



Step 17: No new neighbors from M

Legend



DFS Traversal

Step 18: Process node J

Legend

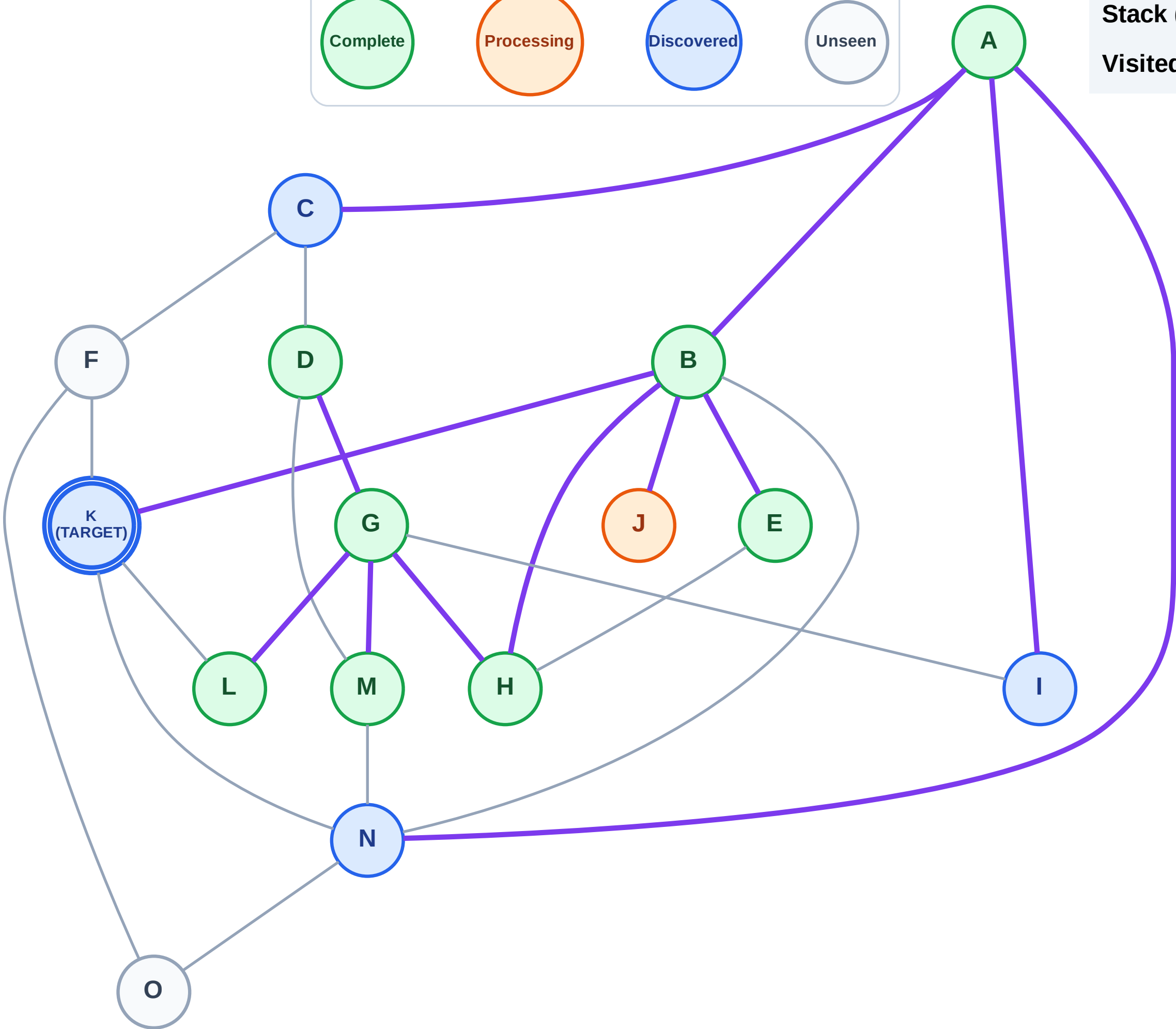
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C | K
Visited: A, B, D, E, G, H, L, M



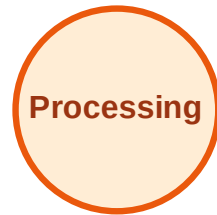
DFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



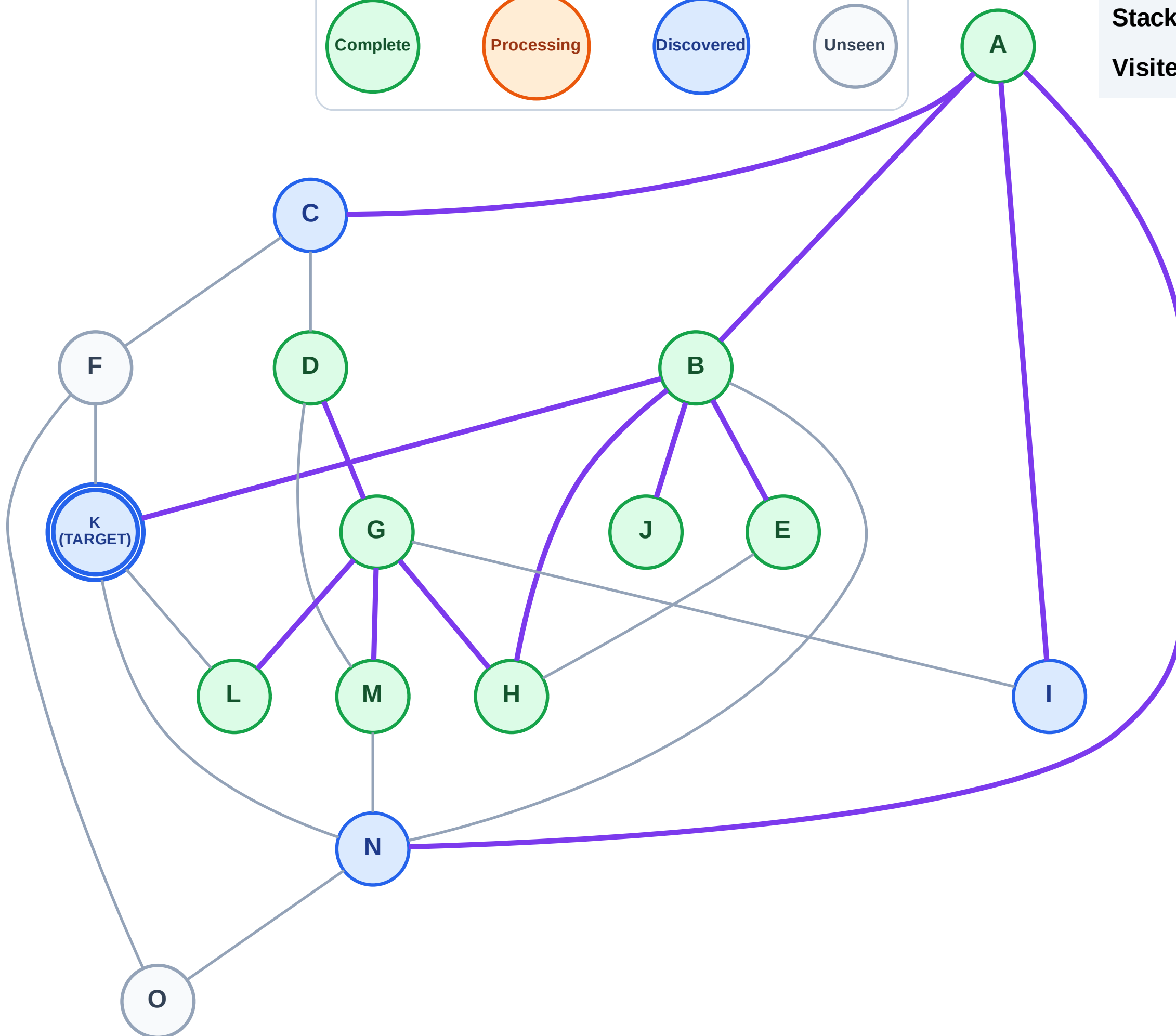
Discovered



Unseen

Stack (top at right): N | I | C | K

Visited: A, B, D, E, G, H, J, L, M



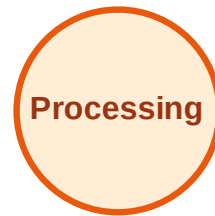
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing

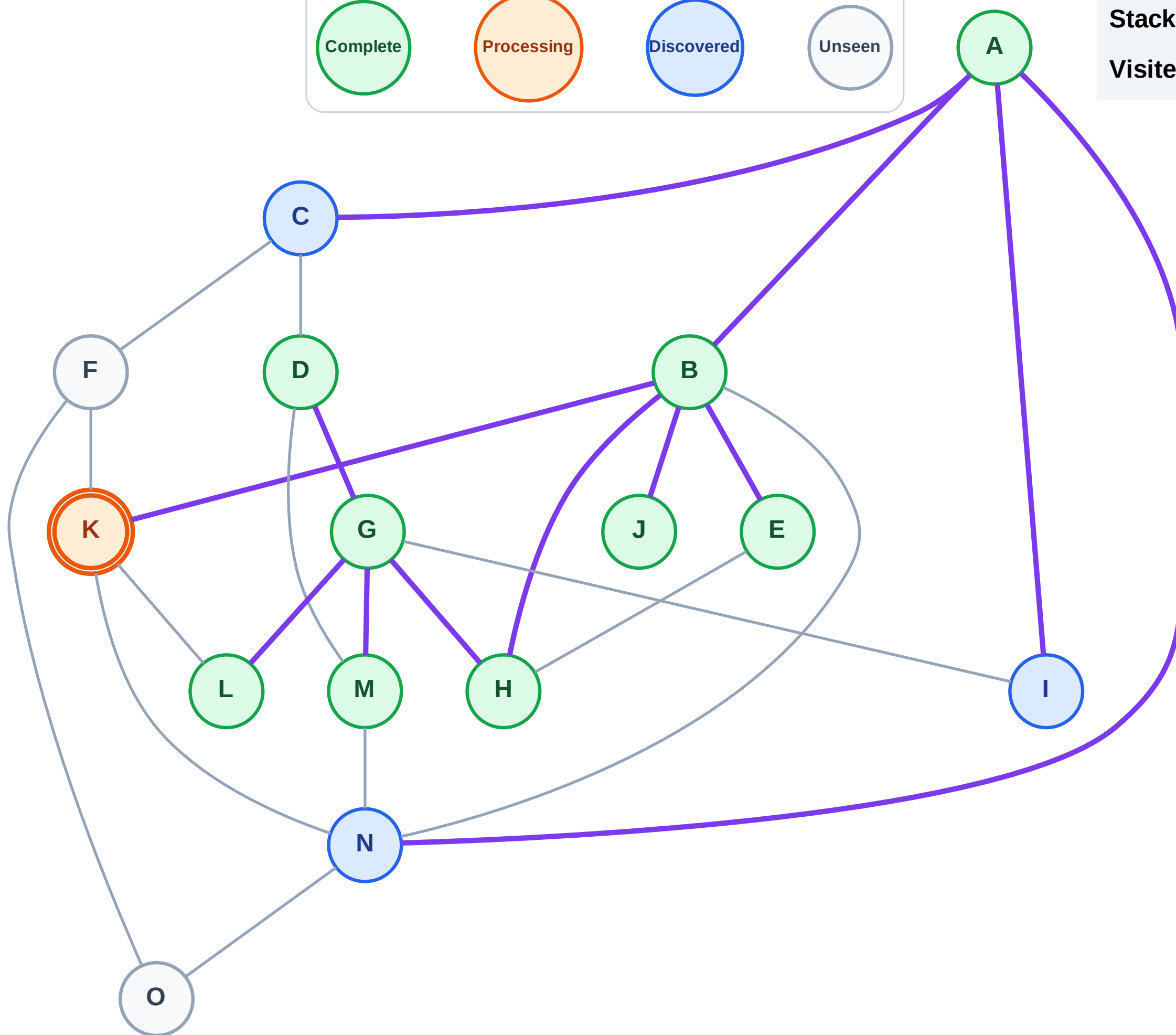


Discovered



Unseen

Stack (top at right): N | I | C
Visited: A, B, D, E, G, H, J, L, M



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C
Visited: A, B, D, E, G, H, J, K, L, M

